
DISCELL

Anonymous Author
Anonymous Institution

Abstract

A cell’s measured transcriptome in targeted in situ spatial transcriptomics mixes three things: what the cell intrinsically is, how its microenvironment has changed it, and transcripts that were assigned to it but belong to its neighbours. The second and third produce the same first-order signature, a cell resembling its neighbours, so no model fit to a single section can tell them apart without further assumptions. We present DISCELL, a variational model that splits a cell’s expression into an intrinsic latent state $\mathbf{z}$, a latent response $\mathbf{w}$ to a deterministic description of its microenvironment, and a fixed leakage mixture over graph neighbours governed by a leak fraction κ . Type-conditional invariance keeps niche-predictable information out of $\mathbf{z}$ through an adversarial regulariser, with a closed-form Gaussian penalty as the cheaper first attempt, and is graded by an independently trained held-out probe rather than by the training objective. Because κ is not identifiable from a single section, we do not estimate it: every model is refitted on a grid of κ values, and the deliverable is which spatial effects survive the grid. On a 407,120-cell Xenium ovarian carcinoma section we show that $\mathbf{z}$ retains within-type intrinsic variation, such as cell-cycle phase, that $\mathbf{w}$ carries the spatially organised variation, and we report which of these readouts are stable across the leakage sweep. **[TODO: abstract numbers and claims to be finalised after the multi-seed reruns]**

1 INTRODUCTION

Imaging-based spatial transcriptomics assays such as Xenium (Janesick et al., 2023) measure thousands of transcripts per cell in situ, at single-cell resolution, over sections containing hundreds of thousands of cells. The appeal is obvious: a cell’s expression can be read together with the identity and state of the cells around it, which is the raw material for asking how a microenvironment shapes the cells within it. The difficulty is that a cell’s measured count vector is not a clean readout of that cell. It is at least three things superimposed. First, the cell’s *intrinsic* state: its type, its cell-cycle phase, its clonal identity, and the systemic signals that reach every cell of its type regardless of position. Second, its *response* to the local microenvironment: the programmes it has switched on or off because of who its neighbours are and what the surrounding tissue architecture looks like. Third, a *measurement artefact* specific to in situ assays: transcripts that were detected in the vicinity of a segmentation boundary and assigned to the wrong cell. Segmentation of densely packed tissue is imperfect, and published estimates place the fraction of misassigned transcripts anywhere between one tenth and one half **[NEEDS A SOURCE]**.

The second and third components are confounded at first order. A cell that responds to its neighbours comes to resemble them; a cell whose counts are contaminated by its neighbours also resembles them. Neighbour composition predicts both. The two differ in form, one adds a neighbour’s own profile, the other shifts the cell’s own programmes, but from the cell-by-gene count matrix alone the split between them is fixed only by the parametric forms one is willing to assume, and a flexible decoder can mimic either. Any claim that a spatial effect on expression is real therefore depends, silently, on how much of the resemblance was attributed to contamination.

This paper makes that dependence explicit rather than hiding it. We propose DISCELL (*disentangled cell states*), a generative model of per-cell counts with two latent variables and one fixed mixing operator. A latent $\mathbf{z}_i$ carries the cell’s intrinsic state and is, by architec-

ture and by regularisation, kept free of information about the microenvironment given the cell’s type. A latent $\mathbf{w}_i$ carries the cell’s response to a deterministic context descriptor $\mathbf{c}_i$ that summarises who and what is nearby. A sparse, precomputed leakage kernel β mixes a fraction κ of each neighbour’s clean expression profile into the cell’s expected composition. The leak fraction κ is not estimated. It is fixed, the model is refitted on a grid of κ values, and every spatial effect is reported as a function of κ with its envelope over random seeds, a sensitivity analysis in which κ is the sensitivity parameter. An effect that is stable across the grid and across seeds is a finding; one that changes sign or vanishes is not separable from leakage on this data. The range across the grid is the deliverable, reported alongside an operating point. It is not a confidence interval, and it does not shrink with more cells.

Our modelling choices follow from three requirements that we motivate in Section 2 and Appendix B. The decoder receives no type label, so that $\mathbf{z}$ is forced to carry identity and cannot degenerate into a residual code. The context encoder queries the neighbourhood with the cell’s type rather than with $\mathbf{z}_i$, closing a channel through which attention would otherwise reconstruct the cell from its most similar neighbours and make the prior on $\mathbf{w}$ ego-informed. And the invariance we enforce is *conditional* on type: cell types cluster in space, so a marginal invariance between $\mathbf{z}$ and neighbourhood composition deletes cell type from $\mathbf{z}$ altogether, a failure mode documented for conditional subspace VAEs (Klys et al., 2018; Slavutsky et al., 2026).

Related work. Node-centric expression models (Fischer et al., 2023) regress a cell’s expression on its neighbourhood directly and produce a niche descriptor without a per-cell latent; our context descriptor $\mathbf{c}_i$ plays that role, and $\mathbf{w}_i$ is what remains after conditioning on it. Spatially informed variational models that split a cell’s state into an intrinsic and a spatially induced part with an independence regulariser (Dong et al., 2025), and niche-characterisation models built on graph neural networks (Birk et al., 2025), share our design goal; none of them models transcript leakage, which is the confound this paper is about [TODO: verify these characterisations against the cited papers; add scVIVA and further spatial VAEs]. Contamination-aware models for in situ data, notably resolVI (Ergen et al., 2025), model transcript misassignment explicitly and estimate a per-cell mixing weight; we instead treat the section-wide leak fraction as un-identified and sweep it. resolVI also disables gradients through the neighbour path because the coupling otherwise collapses the fit; we adopt the same stop-gradient and describe the degeneracy it prevents. Variational disentanglement of condition-specific from shared representations

(Klys et al., 2018; Slavutsky et al., 2026) supplies the two-bound objective and the posterior-regularisation view of an adversarial invariance term (Ganchev et al., 2010) that we build on. Two things differ. The prior on our condition-specific code conditions on a learned, graph-derived context, while the invariance is enforced against fixed data descriptors of the niche, so the model cannot shape the condition to make invariance trivial; and the confound we guard against is a measurement artefact rather than a nuisance covariate. That adversarial removal is not sufficient without an independent check is the observation of Elazar and Goldberg (2018); our probe protocol is its application. [TODO: extend related work: segmentation and contamination correction methods, further spatial VAEs and GNN spatial models]

Contributions. (i) A generative model that decomposes in situ counts into intrinsic state, microenvironmental response and neighbour leakage, with a multinomial likelihood in which the leakage mixture is exact under Poisson sources (Section 2.2). (ii) An inference scheme whose architecture carries most of the identification argument: the intrinsic encoder never sees the context, the attention context is queried by type, and neighbour quantities are frozen, with an exact two-hop batching scheme that makes the frozen path affordable (Sections 2.3 and 2.6). (iii) A conditional-invariance regulariser, adversarial in its operating configuration with a closed-form penalty as the cheaper first attempt, together with a held-out probe protocol that grades either independently of the training objective (Section 2.5). (iv) The leakage sweep as a sensitivity analysis: spatial effects are reported as trajectories over κ with seed envelopes, alongside an operating point (Section 2.8). (v) An evaluation on a 407,120-cell ovarian carcinoma section [PENDING RUNS: second section, multi-seed error bars].

2 THE DISCELL MODEL

We present the model in the order in which its parts depend on one another: the observed quantities and the spatial graph (Section 2.1); the generative model and the assumptions behind it (Section 2.2); the inference networks (Section 2.3); the objective (Section 2.4); the conditional-invariance regulariser and the held-out probe that grades it (Section 2.5); the frozen-neighbour batching that makes the model trainable (Section 2.6); model selection (Section 2.7); and what is and is not identified, which is what turns a fit into a finding through the leakage sweep (Section 2.8). Each design decision is stated with its reason; the longer arguments are in Appendix B and the derivations in Appendix A.

2.1 Setting and Notation

We consider a single tissue section profiled with a targeted in situ assay. For each of N segmented cells i we observe a count vector $\mathbf{x}_i \in \mathbb{N}^G$ over G genes, its total $\ell_i = \sum_g x_{ig}$, a cell-type label $t_i \in \{1, \dots, K\}$, and the morphology image around the cell. The label may be a curated annotation or a frozen unsupervised clustering and is treated as data, with two caveats that recur below: it was itself derived from the same contaminated counts, and its granularity is a modelling choice that sets what the decomposition is relative to (Section 2.5). The total ℓ_i is conditioned on throughout and never modelled.

Spatial graph. Cells are nodes; two cells are adjacent if and only if their Voronoi cells share a face, i.e. no third cell lies between them (Delaunay adjacency). This choice has no neighbourhood count k and no radius, adapts to local packing with an average degree near six, and uses only cell centroids, so it is insensitive to segmentation-boundary noise and to the nuclear-expansion segmentation that in situ platforms fall back on. Delaunay adjacency also connects cells across lumens, vessels and tears, so edges longer than a cut-off ($40 \mu\text{m}$ here, well above the 95th percentile of edge length) are pruned. We write $\mathcal{N}_i$ for the pruned neighbour set of i , which excludes i itself; all its members are first-order neighbours, so no cell-size correction of distances is needed. The same graph serves every role below.

For each edge we record the centroid distance d_{ij} and the length f_{ij} of the shared Voronoi face, measured after clipping each Voronoi cell to a $30 \mu\text{m}$ disc around its centroid so that faces at tissue borders and across gaps are bounded. The face length is a proxy for contact extent that is defined whether or not the segmented polygons touch. We deliberately do not use the length of shared segmented boundary: under nuclear-expansion segmentation, whether two polygons touch is largely a measurement of local density, and local density is a property of the niche that we want to keep out of the *technical* part of the model, the part that describes measurement rather than biology (Appendix B.2).

Derived observables. From the graph and labels we form the neighbour-type composition $\mathbf{y}_i \in \Delta^{K-1}$, $y_{ik} = |\{j \in \mathcal{N}_i : t_j = k\}|/|\mathcal{N}_i|$, which excludes the cell itself. From the image we form an embedding $\Phi_i \in \mathbb{R}^{d_\Phi}$ of a patch centred on the cell, produced by a frozen pretrained vision model, in which a disc covering the cell has been masked out, so that Φ_i describes the surrounding tissue and not the cell (Appendix D.4). The two descriptors act at different scales: the graph is one hop, a few to thirty micrometres, while the

masked patch describes tissue beyond roughly $25 \mu\text{m}$; both are what we mean by “the niche”. Here $d_\Phi = 384$, the embedding’s native width, since an ablation found no benefit from projecting it (Appendix E). Three lookup tables are fixed before training: $\bar{\mathbf{y}}(t)$, the mean composition of cells of type t ; $\bar{\Phi}(t)$, the mean image-niche membership of type t ; and $\bar{\mathbf{v}}(t)$, the mean of the invariance block of Section 2.5 for type t . Isolated cells, those with no neighbours after pruning, have an empty composition and are excluded from all three tables.

Leakage kernel. The leakage kernel is a sparse matrix over the same edges, normalised over the neighbours of each connected cell,

$$\beta_{ij} \propto f_{ij} \exp(-d_{ij}/\tau), \quad \sum_{j \in \mathcal{N}_i} \beta_{ij} = 1, \quad (1)$$

with contamination decay length $\tau = 20 \mu\text{m}$; it is row-stochastic on connected cells, and an isolated cell keeps an all-zero row. It is precomputed once and never learned. At this cell spacing the decay is nearly inert and the kernel is dominated by the face lengths (Appendix D.3); it is retained for platforms with sparser cells. Continuous geometry is allowed here, and only here: β is not part of any conditioning variable and does not touch the estimand (Appendix B.2).

2.2 Generative Model

Each cell has two latent variables. The intrinsic state $\mathbf{z}_i \in \mathbb{R}^{d_z}$ carries what the cell is; the response $\mathbf{w}_i \in \mathbb{R}^{d_w}$ carries what its environment did to it. The environment itself enters as a deterministic context descriptor $\mathbf{c}_i$ (defined in Section 2.3) that is shared by co-located cells. With m_ψ a learned prior-mean network, a a learned decoder, $\mathbf{B} \in \mathbb{R}^{G \times d_w}$ a learned loading matrix and $\kappa \in [0, 1]$ the fixed leak fraction,

$$\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (2)$$

$$\mathbf{w}_i \mid \mathbf{c}_i, t_i \sim \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I}), \quad (3)$$

$$\boldsymbol{\rho}_i = \text{softmax}(a(\mathbf{z}_i) + \mathbf{B}\mathbf{w}_i), \quad (4)$$

$$\bar{\boldsymbol{\rho}}_i = \sum_{j \in \mathcal{N}_i} \beta_{ij} \boldsymbol{\rho}_j, \quad (5)$$

$$\mathbf{p}_i = (1 - \kappa) \boldsymbol{\rho}_i + \kappa \bar{\boldsymbol{\rho}}_i, \quad (6)$$

$$\mathbf{x}_i \mid \ell_i \sim \text{Multinomial}(\ell_i, \mathbf{p}_i). \quad (7)$$

Here $\boldsymbol{\rho}_i \in \Delta^{G-1}$ is the cell’s own *clean* composition, $\bar{\boldsymbol{\rho}}_i$ is the *foreign influx*, the leakage-weighted average of its neighbours’ clean compositions, and $\mathbf{p}_i$ is the fitted multinomial probability. The decoder $a(\mathbf{z}_i) \in \mathbb{R}^G$ is the cell’s baseline programme in log space; column k of $\mathbf{B}$ is a gene programme and w_{ik} is how far cell i has moved along it. The prior variance is fixed at $\sigma_w = 1$, deliberately (Section 2.3).

For an isolated cell $\bar{\rho}_i = \mathbf{0}$, and the mixture in Equation (6) would sum to $1 - \kappa$, charging the cell $-\ell_i \log(1 - \kappa)$ nats for leakage it cannot receive. The mixture is therefore renormalised per row,

$$\mathbf{p}_i = \frac{(1 - \kappa) \boldsymbol{\rho}_i + \kappa \bar{\boldsymbol{\rho}}_i}{(1 - \kappa) + \kappa \sum_{j \in \mathcal{N}_i} \beta_{ij}}, \quad (8)$$

so that an isolated cell degenerates to $\mathbf{p}_i = \boldsymbol{\rho}_i$, i.e. $\kappa_i = 0$: no neighbours, no foreign source. For connected cells the denominator is one. Probabilities are floored at 10^{-8} inside the logarithm. Isolated cells keep an all-zero composition $\mathbf{y}_i$ and are not excluded from the regulariser or the probes of Section 2.5; they are about two in a thousand cells on the primary section (899 of 407,120 after pruning).

What the leakage term assumes. The leak fraction κ is section-wide, not per cell, and there is no ambient term. A cell’s contamination in reality varies with its size, its library size and the abundance of its neighbours; that per-cell variation is absorbed by $\mathbf{z}$ and $\mathbf{w}$ and is not addressed by the sweep. Diffuse background is assumed negligible relative to leakage from adjacent cells [NEEDS A SOURCE]. The mixture in Equation (6) is exact under two assumptions: that the sources are independent Poisson variables, whose sum is again Poisson, and that the rate of leaked-in transcripts scales with the receiving cell’s own rate rather than with the donors’, which is where the per-cell variation just mentioned is hidden. The multinomial then follows from conditioning on the observed total (below).

What the decomposition means. The split between $\mathbf{z}$ and $\mathbf{w}$ is defined by two requirements. First, within a type, the intrinsic state should carry nothing that is predictable from the niche: $\mathbf{I}(\mathbf{z}_i; [\mathbf{y}_i, \boldsymbol{\Phi}_i] \mid t_i) = 0$. In Equation (2) $\mathbf{z}_i$ is a priori independent of everything, and $\mathbf{y}_i, \boldsymbol{\Phi}_i$ are not modelled, so this is not a property of the generative model; it is a constraint imposed on the variational family in Section 2.5, in the sense of posterior regularisation (Ganchev et al., 2010). Second, the environment reaches the counts only through the response and through leakage: $\mathbf{x}_i \perp \mathbf{c}_i \mid \mathbf{z}_i, \mathbf{w}_i, \{\boldsymbol{\rho}_j\}_{j \in \mathcal{N}_i}$, which holds by construction because the decoder of Equation (4) takes no context. Precisely, what the first requirement removes from $\mathbf{z}$ is variation predictable from the neighbour composition and the masked image given the type. It does not remove intrinsic variation that happens to be spatially organised within a type, nor variation carried by the neighbours’ own intrinsic codes; on a carcinoma section, where subclonal structure is the dominant within-type spatial variation, this is the main case rather than a corner case, and it is why the decomposition is relative to t (Section 2.5).

Why a multinomial. If $x_{ig} \sim \text{Poisson}(\lambda_{ig})$ independently, then $\mathbf{x}_i \mid \sum_g x_{ig} = \ell_i$ is $\text{Multinomial}(\ell_i, \boldsymbol{\lambda}_i / \sum_g \lambda_{ig})$. Since ℓ_i is observed, the multinomial is the coherent likelihood: no library-size latent and no double counting of depth. Poisson sources are closed under addition, which is what makes the leakage mixture exact; negative-binomial sources with gene-specific means are not in general, and a negative binomial on the mixed counts would let the dispersion absorb part of what the mixture should explain. At the depth of in situ data, tens to a few hundred counts over thousands of genes, gene-wise overdispersion cannot be estimated in any case, and a low-dimensional amortised $\mathbf{z}$ absorbs much of what a dispersion parameter captures in coarser models. Should posterior predictive checks show overdispersion at higher depth, the appropriate extension is a Dirichlet-multinomial with one concentration scalar, not G dispersions.

Why κ is fixed. Leakage and response differ in form: one adds a neighbour’s own profile through a fixed contact kernel, the other shifts the cell’s own programmes along learned directions. From the cell-by-gene count matrix alone, however, the split between them is pinned only by those parametric forms, and a decoder flexible enough to fit the data can mimic either, since both make a cell resemble its neighbours and neighbour composition predicts both. We do not trust the likelihood to choose κ . Instead we fix it, fit the model independently at every point of a grid, and report each readout as a function of κ (Section 2.8). $\kappa = 0$ is the nested no-correction null. The grid is anchored at zero and extends to 0.4, close to the upper range of published misassignment estimates for this platform [NEEDS A SOURCE]; a data-driven upper bound from an atlas-defined marker set, and the transcript-level information that would identify κ from a nuclear/extranuclear split of the counts, are discussed in Appendix B.5.

Why the decoder has no type input. The decoder in Equation (4) takes $\mathbf{z}_i$ and $\mathbf{w}_i$ only, so the only route from identity to expression runs through $\mathbf{z}$. If $\mathbf{z}$ collapsed to a function of the label, both reconstruction terms in Section 2.4 would lose all within-type variance, a large fitting penalty, and that is what holds $\mathbf{z}$ open. With the label in the decoder, $\mathbf{z}$ would become a residual-only code and would no longer be interpretable as identity. What $\mathbf{z}$ carries beyond the label is what the label discretises away: cell-cycle phase, clonal identity, position on a differentiation continuum, stress and metabolic state, states acquired in a niche the cell has since left, and systemic signals that affect every cell of a type regardless of position. The last is environmental in the colloquial sense but not niche-predictable, and it belongs in $\mathbf{z}$. Two remarks follow. Since the encoder of

Table 1: Learned and fixed quantities. $G = 5,101$, $K = 18$, $d_z = 20$, $d_w = 6$, $d_\Phi = 384$ on the primary section.

Learned	Fixed before training
encoder of $q(\mathbf{z}_i \mathbf{x}_i, t_i)$; attention layer and type query embed(t); prior mean m_ψ ; encoder of $q(\mathbf{w}_i \cdot)$; decoder a ; loadings $\mathbf{B}$; adversary heads (separate optimiser)	graph, kernel β , τ ; leak fraction κ (swept); prior scale $\sigma_w = 1$; image embedding Φ and its principal components; image-niche memberships e^Φ ; tables $\bar{\mathbf{y}}(t)$, $\bar{\Phi}(t)$, $\bar{\mathbf{v}}(t)$; labels t ; totals ℓ

Section 2.3 receives the label, the agreement between $\mathbf{z}$ and t measures whether $\mathbf{z}$ *retains* type, not whether it discovers it; its collapse is the detector for the marginal-invariance failure mode (Section 2.7). And coarse labels leave more for $\mathbf{z}$ to carry; whether $\mathbf{z}$ separates cell-cycle phases within a type is checked in Section 3.5.

2.3 Inference Networks

Table 1 summarises what is learned and what is fixed. All networks are multilayer perceptrons unless stated; widths are in Table 2.

The context descriptor $\mathbf{c}_i$. One vector per cell summarising who and what is nearby. It is deterministic, is shared by co-located cells, and is the conditioning variable of the model. It is computed by a single graph-attention layer (Brody et al., 2022) over the pruned graph, concatenated with the image embedding and an isolation flag:

$$\mathbf{h}_j = [\text{onehot}(t_j), \text{sg}[\boldsymbol{\mu}_z(\mathbf{x}_j, t_j)]], \quad (9)$$

$$\mathbf{g}_i = \text{GATv2}(\text{query} = \text{embed}(t_i), \{\mathbf{h}_j\}_{j \in \mathcal{N}_i}), \quad (10)$$

$$\mathbf{c}_i = \mathbf{g}_i \oplus \Phi_i \oplus \mathbf{1}[\mathcal{N}_i = \emptyset]. \quad (11)$$

Here $\boldsymbol{\mu}_z(\mathbf{x}_j, t_j)$ is the posterior mean of $\mathbf{z}_j$ (below), $\text{sg}[\cdot]$ denotes a stop-gradient, and $\text{embed}(t_i)$ is a learned per-type vector of the same width as the source features. The layer has no self-loops, no edge features, and its heads are averaged. Its attention weights π_{ij} satisfy $\sum_j \pi_{ij} = 1$. Three decisions here are load-bearing.

The query is the type, not $\mathbf{z}_i$. In attention of this form the destination feature enters the attention logits but never the aggregated value, so using $\mathbf{z}_i$ as the query would inject no ego content directly. It would, however, open a selection channel: attention can learn $\pi_{ij} \propto \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j))$ and reconstruct the cell through its most similar neighbours, so that $\mathbf{c}_i$ becomes a mirror of $\mathbf{z}_i$. The incentive exists because an ego-informed prior on $\mathbf{w}_i$ is free decoder capacity, and it is strongest in homotypic regions, where κ is also least identified. The consequence would be that m_ψ already knows the

cell, the divergence $\text{KL}(q(\mathbf{w}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i))$ collapses, and $\mathbf{w}_i$ measures deviation from *self* rather than from the expected response. Querying with the type closes the selection channel at no cost to the prior, since $m_\psi(\mathbf{c}_i, t_i)$ already receives t_i . A value channel remains: in homotypic regions the neighbours’ codes resemble the cell’s through shared lineage and through symmetric contamination, so $\mathbf{c}_i$ can correlate with $\mathbf{z}_i$ regardless of the query. That residual is monitored, not assumed away (Section 2.7; Appendix B.1).

No per-edge geometry enters the context. The graph says who is adjacent; the image describes how packed and how structured the region is; per-edge distances and contact extents are withheld from $\mathbf{c}_i$. Three reasons, in decreasing order of weight. The concrete risk is collinearity: an attention layer given edge features can rediscover the leakage kernel β , after which $\mathbf{w}$ becomes collinear with $\bar{\rho}$ and the two mechanisms the sweep is meant to separate merge inside the model (Appendix B.2). Within a first-order neighbourhood the distances span perhaps 5 to 30 μm , most of which is local density, which the image already carries. And the intended reading of the context is that neighbour-type composition is the exposure of interest while the image and the graph are conditioning covariates; a hypothetical change in composition is a meaningful question about a tissue in a way that “the same neighbours, further away” is not, and putting per-edge geometry into the conditioning variable would mix the two. The estimand is used only informally here; counterfactual machinery is future work (Section 4).

Isolated cells. A cell with no in-edges has an empty attention set; the graph part of $\mathbf{c}_i$ is set to zero and the flag is raised. The image part stays, since Φ_i is measured data defined for every cell. Together with Equation (8), an isolated cell is modelled with $\kappa_i = 0$ and an image-only context.

Posterior over $\mathbf{z}$. The posterior $q(\mathbf{z}_i | \mathbf{x}_i, t_i)$ is a diagonal Gaussian produced by a network whose input is the library-normalised, log-transformed count vector together with the log total and the one-hot type,

$$[\log(1 + \mathbf{x}_i \tilde{\ell} / \ell_i), \log \ell_i, \text{onehot}(t_i)], \quad \tilde{\ell} = \text{median}_i \ell_i. \quad (12)$$

The encoder *never sees* $\mathbf{c}_i$: invariance by architecture, which the regulariser of Section 2.5 then completes. Library normalisation before the log matters because a raw $\log(1 + \mathbf{x})$ entangles depth with composition, so that a 300-count and a 50-count cell of the same type look very different; since ℓ_i is conditioned on rather than modelled, $\mathbf{z}$ should not spend capacity on it. Total counts are nonetheless weakly informative about state, so $\log \ell_i$ enters as one extra scalar rather than being discarded.

Prior and posterior over $\mathbf{w}$. The prior is what a typical cell of this type would do in this context; the posterior is what this cell did:

$$p(\mathbf{w}_i | \mathbf{c}_i, t_i) = \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I}), \quad (13)$$

$$q(\mathbf{w}_i | \mathbf{c}_i, t_i, \mathbf{z}_i, \mathbf{x}_i) = \mathcal{N}(\boldsymbol{\mu}_{w,i}, \text{diag } \boldsymbol{\nu}_{w,i}^2), \quad (14)$$

with $\boldsymbol{\nu}_{w,i}$ the learned posterior scale, as distinct from the fixed prior scale σ_w . Both distributions are functions of the context, but only the posterior sees the cell itself, through the sampled $\mathbf{z}_i$ and through the encoded counts of Equation (12); we abbreviate the posterior as $q(\mathbf{w}_i)$ where its arguments are clear. The divergence $\text{KL}(q \| p)$ reads as how far this cell deviates from its expected response, an anomaly score that the objective provides at no extra cost; whether it carries usable per-cell information at the operating point is an empirical question, reported in Section 3.6. The prior is also what would make counterfactuals possible, by drawing $\mathbf{w}' \sim p(\mathbf{w} | \mathbf{c}', t)$ and decoding, with abduction $\mathbf{w}'_i = m_\psi(\mathbf{c}', t_i) + \sigma_w \hat{\boldsymbol{\epsilon}}_i$ carrying the cell's own residual $\hat{\boldsymbol{\epsilon}}_i = (\mathbf{w}_i - m_\psi(\mathbf{c}_i, t_i))/\sigma_w$; that machinery is not exercised in this paper.

The direct path from $\mathbf{x}_i$ into the posterior is necessary. The conditional independence $\mathbf{w} \perp \mathbf{x} | \mathbf{z}, \mathbf{c}$ is false by explaining-away: $\mathbf{x}$ is generated from both $\mathbf{z}$ and $\mathbf{w}$, so knowing $\mathbf{x}$ and $\mathbf{z}$ informs $\mathbf{w}$ directly, which is also why Slavutsky et al. (2026) condition their condition-specific posterior on the data rather than on the shared code. In our setting $\mathbf{z}$ is lossy in exactly the wrong direction: it is low-dimensional, the invariance regulariser removes niche-predictable variation from it, and that variation *is* the spatial response that $\mathbf{w}$ exists to capture. Without the direct path, $\mathbf{w}_i$ collapses to $\mathbb{E}[\text{response} | \text{type}, \text{niche}]$, identical for co-located similar cells, and the divergence vanishes (Appendix B.3).

Learned and fixed variances. Both posteriors carry learned diagonal variances; this is what makes the reparameterised sample meaningful (Kingma and Welling, 2014; Rezende et al., 2014) and lets uncertainty vary per cell, so that a 30-count cell can have a wider $q(\mathbf{z})$ than a 300-count cell. The prior scale σ_w is fixed at one, for two reasons of which the first is decisive. A learned *constant* σ_w is a gauge freedom: $\mathbf{w}$ and $\mathbf{B}$ enter the decoder only as $\mathbf{B}\mathbf{w}_i$, so rescaling dimension k of $\mathbf{w}$ by s and column k of $\mathbf{B}$ by $1/s$ leaves the likelihood unchanged, and a learned constant prior scale would absorb the compensating change in the prior; fixing it pins the scale. A learned *conditional* $\sigma_w(\mathbf{c}, t)$ would be meaningful, claiming that some niches produce more variable responses than others, but it opens a degenerate direction: the model can shrink $\text{KL}(q \| p)$ for free by inflating $\sigma_w(\mathbf{c}, t)$ wherever the penalty bites, making the prior vague rather than making $\mathbf{w}$ predictable. If it is wanted, it must be bounded (Appendix B.6).

Decoder. The decoder is Equation (4): a network $a(\mathbf{z}_i) \in \mathbb{R}^G$ plus the linear shift $\mathbf{B}\mathbf{w}_i$, followed by a softmax over genes. It takes no type input, for the reason given in Section 2.2. If reconstruction is poor the remedy is to widen $\mathbf{z}$ or the network, not to add t .

2.4 Objective

With the neighbour quantities $\mathbf{c}_i$ and $\bar{\rho}_i$ held constant (Section 2.6), the evidence lower bound on $\log p(\mathbf{x}_i | \mathbf{c}_i, t_i)$ under the variational family $q(\mathbf{z}_i) q(\mathbf{w}_i | \mathbf{c}_i, t_i, \mathbf{z}_i, \mathbf{x}_i)$ is

$$\begin{aligned} L_i = \mathbb{E}_{q(\mathbf{z}_i)} \Big[& \mathbb{E}_{q(\mathbf{w}_i | \mathbf{z}_i)} [\log p(\mathbf{x}_i | \mathbf{z}_i, \mathbf{w}_i)] \\ & - \text{KL}(q(\mathbf{w}_i | \mathbf{z}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i)) \Big] \\ & - \text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i)), \end{aligned} \quad (15)$$

where the $\mathbf{w}$ -divergence sits inside the outer expectation because $q(\mathbf{w}_i | \cdot)$ conditions on the sampled $\mathbf{z}_i$; evaluating the closed-form Gaussian divergence at the one reparameterised draw is an unbiased one-sample estimate of it (Appendix A.1). A second bound on the same data is obtained for the model in which $\mathbf{w}$ is not inferred, i.e. $q(\mathbf{w}_i) := p(\mathbf{w}_i | \mathbf{c}_i, t_i)$, which zeroes that divergence:

$$\begin{aligned} L_i^z = \mathbb{E}_{q(\mathbf{z}_i)} \mathbb{E}_{p(\mathbf{w}_i | \mathbf{c}_i, t_i)} \Big[& \log p(\mathbf{x}_i | \mathbf{z}_i, \mathbf{w}_i) \\ & - \text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i)) \Big]. \end{aligned} \quad (16)$$

We call this the *intrinsic path*, and it is load-bearing. The first reconstruction term does reward $\mathbf{z}$ for carrying signal, since $a(\mathbf{z})$ is the only route to all G genes, but with the counts also wired into $q(\mathbf{w})$ the allocation between $\mathbf{z}$ and $\mathbf{w}$ is ambiguous, and the second bound breaks the tie toward $\mathbf{z}$: under it $\mathbf{z}$ must explain the cell given only the *expected* response. Because it runs through the same leakage mixture as the first bound, it never asserts $\kappa = 0$ and so never contradicts it (Appendix A.2).

The training objective sums the two bounds, with a weight ω on the intrinsic path and per-term weights on the divergences,

$$\begin{aligned} J = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \Big[& \frac{1}{\ell_i} \sum_g x_{ig} \log p_{ig}(\mathbf{z}_i, \mathbf{w}_i) \\ & + \omega \frac{1}{\ell_i} \sum_g x_{ig} \log p_{ig}(\mathbf{z}_i, \tilde{\mathbf{w}}_i) \\ & - (1 + \omega) \alpha_z \text{KL}(q(\mathbf{z}_i) \| \mathcal{N}(\mathbf{0}, \mathbf{I})) \\ & - \alpha_w \text{KL}(q(\mathbf{w}_i) \| \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I})) \Big] \\ & - \alpha_a \text{Pen}, \end{aligned} \quad (17)$$

where $\mathcal{S}$ is the set of cells whose loss is evaluated in a batch, $\tilde{\mathbf{w}}_i \sim p(\mathbf{w}_i | \mathbf{c}_i, t_i)$ is a draw from the prior, and

Pen is the invariance term of Section 2.5, a functional of the whole batch; under the adversarial escalation it is replaced by the per-cell mean $\frac{1}{|\mathcal{S}|} \sum_i \text{Adv}_i$. One decoder is evaluated twice, with $\mathbf{w}$ from the posterior in the first reconstruction term and from the prior in the second. The factor $(1+\omega)$ is what keeps J proportional to $L_i + \omega L_i^z$: each of Equations (15) and (16) carries its own $-\text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i))$, so their weighted sum contains $(1+\omega)$ copies of it, and omitting the second copy, the natural mistake, breaks the correspondence even at unit weights.

Scaling. The reconstruction terms are divided by ℓ_i . Unscaled, they are of order -200 to -400 nats per cell against divergences of order 1 to 10, and their magnitude varies severalfold between sections, so weights tuned on one section would not transfer to the next. The scaling changes what the weights mean: reconstruction becomes $O(1)$ while the divergences stay absolute, so $\alpha_z = 1$ prices a latent’s information roughly ℓ -fold above the unscaled bound and collapses $\mathbf{z}$. At $\alpha_z = \alpha_w = 1/\bar{\ell}$, with $\bar{\ell}$ the mean total count, J is proportional to $L_i + \omega L_i^z$ for a cell of average depth, and that point, not 1, is the centre of any sweep over the weights. A per-cell scaling against global weights also means that a low-count cell’s divergences are weighted more heavily relative to its likelihood than a high-count cell’s, pulling sparse cells toward the prior; we accept this as the price of transferable weights. Once any weight departs from $1/\bar{\ell}$, or through the per-cell scaling itself, J is a weighted surrogate rather than a bound; and because $\mathbf{x}_i$ appears in both reconstruction terms, q is not the posterior of any single model even at the bound-equivalent weights. The objective is coherent as an M-estimation target, and uncertainty in the reported effects comes from the leakage sweep, not from q . The hyperparameters are therefore ω , α_z , α_w and the response width d_w , plus α_a and, under escalation, the adversary’s update schedule; κ is not a hyperparameter but the sweep axis.

2.5 Conditional Invariance of $\mathbf{z}$

Architecture alone does not keep the environment out of $\mathbf{z}$. The counts $\mathbf{x}_i$ carry environmental signal, and with $d_z > d_w$ the intrinsic latent will absorb it given the chance. The target is

$$\mathbb{I}(\mathbf{z}_i; [\mathbf{y}_i, \Phi_i] \mid t_i) = 0. \quad (18)$$

Two features of this target are deliberate. It is *conditional on type*. Types cluster in space, so t_i and $\mathbf{y}_i$ are strongly dependent, and a marginal constraint $\mathbb{I}(\mathbf{z}; \mathbf{y}) = 0$ deletes cell type from $\mathbf{z}$, because carrying “I am a T cell” predicts “my neighbours are T cells”. This is the documented failure of conditional

subspace VAEs (Klys et al., 2018), whose invariance drives the nuisance information to zero while collapsing the code’s agreement with cell type (Slavutsky et al., 2026) [TODO: verify the numbers quoted in the design notes against the DisCoVR paper]. The cost of conditioning is that intrinsic variation that is spatially organised within a type is pushed into $\mathbf{w}$, tumour subclones being the sharp case; the decomposition is relative to t , not absolute, and coarse labels are preferable to fine ones. It also targets *both* $\mathbf{y}$ and Φ . If only composition were penalised, $\mathbf{z}$ would be free to absorb the morphology-defined niche (vessel proximity, fibrosis, density, necrosis), and $\mathbf{w}$ would under-report morphology-driven effects; $\mathbf{y}$ and Φ are correlated, so the part that matters is the residual that is image-predictable but not composition-predictable, and that residual is what would go unpenalised. Neither the penalty below nor the adversary is part of any bound; both are posterior regularisation (Ganchev et al., 2010), a constraint on the variational family written as a penalty, which is the status the adversarial term also has in Slavutsky et al. (2026) (Appendix A.3).

Closed-form penalty. The first regulariser is the Gaussian estimate of the conditional mutual information, which needs no second optimiser. With $\mathbf{v}_i = [\mathbf{y}_i^{(-K)}, \text{PC}(\Phi_i)]$, where $\mathbf{y}_i^{(-K)}$ drops one composition column (the simplex constraint makes the full block singular) and $\text{PC}(\Phi)$ keeps the leading principal components of the image embedding,

$$\text{Pen} = \sum_t p(t) \cdot \frac{1}{2} \left[\log \det \hat{\Sigma}_{\mathbf{v}|t} + \log \det \hat{\Sigma}_{\mathbf{z}|t} - \log \det \hat{\Sigma}_{[\mathbf{z}, \mathbf{v}]|t} \right], \quad (19)$$

evaluated on the posterior means $\boldsymbol{\mu}_z$ rather than on samples, since encoder noise would dilute the measured dependence and hiding dependence behind noise is exactly what the penalty must not permit (Appendix A.4). Three estimation details are not optional at realistic batch sizes. The image block is kept small (a dozen principal components), which keeps the joint block well conditioned and retains the components that carry the niche signal. The per-type moments are accumulated as exponential moving averages across batches (rate 0.05, an effective window of about twenty batches), because a spatial batch contains few cells of any rare type; a type enters the penalty only once its window-weighted cell count reaches ten times the joint dimension, and the fraction of cells excluded by this support floor is reported, since those cells go unregularised. The type weights $p(t)$ are the section-wide type frequencies and are not renormalised after an exclusion. Regularisation shrinks toward the diagonal (Ledoit and Wolf, 2004; Schäfer and Strimmer, 2005),

$\hat{\Sigma} \leftarrow (1-\gamma)\hat{\Sigma} + \gamma \text{diag}(\hat{\Sigma})$, which damps only the correlations; an additive ridge $+\lambda\mathbf{I}$ would inflate every variance by an amount that depends on how ill-conditioned the estimate was, biasing the log-determinant upward most for rare types and weakening the penalty exactly where data are thinnest. Two further details concern the gradient. The value of Equation (19) is taken from the moving-average moments but its gradient from the current batch at full scale, a straight-through estimator (Bengio et al., 2013); otherwise the only differentiable path runs through the averaging weight, and the effective strength of α_a would silently scale with the averaging constant. And the log-determinants are computed on correlation rather than covariance matrices: mutual information is invariant to per-dimension rescaling, so the value is identical, while the gradient of a log-determinant, which involves the inverse matrix, stays bounded when a dimension has near-zero variance, as the composition of cells deep inside one niche does. A jitter of $10^{-5}\mathbf{I}$ is added to the unit-diagonal correlation matrix before the factorisation; on a correlation matrix this is a uniform relative perturbation, not the variance-dependent ridge rejected above.

Adversarial escalation. The Gaussian penalty removes linear dependence only. When the probe below finds that dependence survives it, the penalty is replaced by an adversary (Louizos et al., 2016; Xie et al., 2017; Ganin et al., 2016): two heads with parameters $\xi, \hat{\mathbf{y}}_\xi, \hat{e}_\xi^\Phi : (\mu_{z,i}, t_i) \mapsto \Delta^{K-1}, \Delta^{E_\Phi-1}$, trained by a separate optimiser to predict the composition and an image-niche membership from the posterior mean of the intrinsic code (evaluated on the mean for the same reason as Equation (19)), and an encoder term

$$\begin{aligned} \text{Adv}_i = & \text{CE}(\mathbf{y}_i, \bar{\mathbf{y}}(t_i)) - \text{CE}(\mathbf{y}_i, \hat{\mathbf{y}}_\xi(\mu_{z,i}, t_i)) \\ & + \text{CE}(e_i^\Phi, \bar{\Phi}(t_i)) - \text{CE}(e_i^\Phi, \hat{e}_\xi^\Phi(\mu_{z,i}, t_i)), \end{aligned} \quad (20)$$

which is the heads’ excess predictive skill over knowing the type alone, in categorical cross-entropy. The adversary maximises it, the encoder minimises it, and it is zero at the optimum; the two halves are reported separately. The image target $e_i^\Phi \in \Delta^{E_\Phi-1}$ is a soft cluster membership of Φ_i over E_Φ image niches, obtained by k -means on its leading principal components (never on the raw high-dimensional embedding, where distances are noise-dominated), fitted once and frozen. We set $E_\Phi = K$ so that the two heads face comparable difficulty and one weight α_a serves both; soft rather than hard assignments give a smoother target with no boundary artefacts. The adversary and the probe below therefore target the image through different summaries, discrete memberships and continuous components respectively; the memberships are the natural target for a classification head, the components for the Gaussian penalty and its probe. A weak adversary

is worse than none: heads that are updated too slowly are defeated by the encoder while an independent probe still finds the leak, so head strength must be verified by the probe, never by the heads’ own training loss (Appendix E). This is the observation of Elazar and Goldberg (2018) that adversarial removal is not enough without an independent check.

The probe, and how to decide. Whether the regulariser worked is decided by a *fresh* predictor of the invariance block $\mathbf{v}_i$ from $(\mu_{z,i}, t_i)$, trained on a spatially separate training split and evaluated on held-out cells: never the training adversary, and never the training cells. We report the excess held-out skill over the type-only baseline,

$$\Delta\text{CE} = \frac{1}{d_v} \left(\mathbb{E} \|\mathbf{v} - \bar{\mathbf{v}}(t)\|^2 - \mathbb{E} \|\mathbf{v} - \hat{\mathbf{v}}_{\text{probe}}(\mu_z, t)\|^2 \right), \quad (21)$$

the Gaussian form of the cross-entropy gain on the continuous block (mean squared error over its $d_v = K - 1 + 12$ components, which is the cross-entropy up to constants), with two reference scales that make it interpretable: the *noise floor*, the value obtained when μ_z is permuted across cells within type, and the *uncontrolled baseline*, the value from a model fit at $\alpha_a = 0$. The probe is fitted in two families, a ridge regression and a small multilayer perceptron (Table 2); the permutation floor is reported for the ridge probe, and the decision is taken on the nonlinear one. In our experiments the linear probe sits at the floor while the nonlinear probe still reads a substantial fraction of the uncontrolled leak (Appendix E). The rule, as applied, is to escalate from Equation (19) to Equation (20) when the nonlinear probe of a converged closed-form fit is clearly above the floor, and then to take the smallest α_a that brings the nonlinear probe within a small fraction of the uncontrolled baseline without a loss in the agreement between $\mathbf{z}$ and cell type or in held-out reconstruction. The agreement is the normalised mutual information (Strehl and Ghosh, 2002) between t and a k -means clustering of μ_z with $k = K$ clusters. If no such α_a exists, the invariance is costing cell type and the label set is too fine. **[TODO: state the fractions actually used once the calibration scans are repeated]**

2.6 Frozen Neighbours and Batching

Neighbour quantities are treated as data, not as things to back-propagate through. The neighbour codes $\mu_z(\mathbf{x}_j, t_j)$ in Equation (9) and the neighbour compositions ρ_j in Equation (5) both enter under a stop-gradient. The mechanism this prevents is a degeneracy rather than an explosion: with the path open, cell j ’s clean composition is trained to absorb cell i ’s residual through ρ_i , so ρ_j stops being j ’s own profile, and the

coupling feeds back through $\bar{\rho}_j$. Ergen et al. (2025) disable gradients on precisely this path after observing mode collapse with it enabled; a first-order feedback argument bounds the amplification through the coupling by $1/(1 - \kappa)$ (Appendix A.5). The neighbours' responses $\mathbf{w}_j$ needed for ρ_j are posterior draws computed in the same batched pass and likewise placed under a stop-gradient.

A tempting shortcut is to compute the foreign influx from the baseline programme alone, $\rho_j \approx \text{softmax}(a(\mathbf{z}_j))$, which avoids the neighbours' context entirely. It errs in the direction that matters most. Neighbours share niches, so $\mathbf{w}_j \approx \mathbf{w}_i$; dropping $\mathbf{w}_j$ strips the local spatial programme out of $\bar{\rho}_i$, and whatever the leakage term then fails to absorb is explained by $\mathbf{w}_i$ instead, *inflating* apparent spatial effects, the exact failure the leakage sweep exists to guard against.

Two hops, one pass. Because $\bar{\rho}_i$ needs ρ_j , hence $\mathbf{w}_j$, hence $\mathbf{c}_j$, hence $\mathcal{N}_j$, a training step for cell i reaches two graph hops even though the attention layer itself reaches one. Batching by *contiguous spatial tile* makes this cheap, because the halo of extra cells around a tile is shared rather than disjoint. A tile's cells are the *seeds*; their graph neighbours form *ring one*, for which the context, the response posterior and the clean composition are all computed; the neighbours of ring one form *ring two*, for which only μ_z is computed to serve as attention sources. Both rings are built by look-ups in the same edge list that β uses, never by a distance query: a cell across a pruned edge is not in ring one even if a radius search would return it, and if the halo and the leak kernel disagree the foreign influx is silently wrong. Attention edges are restricted to destinations in the seeds and ring one, since a ring-two cell's context would normalise over a truncated neighbourhood. The loss is evaluated on seeds only. With tiles of a few thousand cells the halo adds roughly a tenth to a fifth to the number of nodes processed per step, falling with tile size (measured, Appendix C.1), cheaper than a cache of clean compositions that must be kept current; scattered seeds, by contrast, would cost many times more nodes than a contiguous tile, so spacing seeds apart is the expensive option, not the cheap one. The arithmetic of a pass is dominated by the encoder and decoder layers, at $O(Ngh)$ for hidden width h ; the foreign influx costs $O(|E|G)$ for $|E|$ directed edges, about $6NG$ here, though its gathered edge-by-gene rows are the largest intermediate tensor of a step. One consequence of tile batching should be stated: the ring of a training tile can contain cells of a held-out tile, whose counts then enter training as inputs under a stop-gradient. No gradient flows from them and no loss is evaluated on them, but they are used transductively **[PENDING RUNS: quantify**

the fraction of training seeds with a held-out neighbour, or add a one-hop buffer around held-out tiles].

2.7 Model Selection and Diagnostics

A model that explains everything with leakage reconstructs held-out cells well, so held-out reconstruction alone is not a valid stopping criterion. We stop on held-out reconstruction and on the agreement between $\mathbf{z}$ and cell type *jointly*: a checkpoint is accepted only if it improves held-out reconstruction while keeping the agreement (computed on a random subsample of all cells) at or above a fixed fraction (0.9) of its running maximum. Held-out cells are entire spatial tiles, never random cells, because spatial autocorrelation leaks the answer through neighbouring cells landing on both sides of a random split (Roberts et al., 2017).

Five diagnostics accompany every fit. (i) The within-type coefficient of determination of $\mathbf{z}_i$ regressed on $\mathbf{c}_i$, against a control in which $\mathbf{c}_i$ is permuted among cells of the same type; a pooled version would partly measure type separability, which both variables legitimately carry, and the within-type form is blind to it. This is the check on the value channel of the mirror described in Section 2.3. (ii) $\text{KL}(q(\mathbf{w}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i))$ per dimension. Its collapse to zero has three causes that the training curves alone cannot distinguish: posterior collapse under a large α_w , the mirror, and a response posterior starved of ego evidence. Each has its own check and fix (Appendix B.3), and the mirror must be ruled out before any free-bits style remedy (Kingma et al., 2016) is applied, since free bits would mask it and leave the anomaly score meaningless. (iii) The held-out probe of Equation (21). (iv) The $\mathbf{z}$ -type agreement, the hard floor whose collapse is the marginal-invariance failure mode. (v) Two degeneracy checks in the opposite direction, asking whether $\mathbf{z}$ has collapsed onto the label: the ratio $I(\mathbf{z}; t)/H(t)$ together with the within-type variance of μ_z , and the gap in held-out reconstruction between $\mathbf{z}$ and a one-hot type code **[PENDING RUNS: compute for the canonical fits]**.

2.8 What Is Identified, and the Leakage Sweep

Identifiability. Four statements can be made. The leak fraction κ is not identified from the count matrix under the parametric forms we are willing to assume, and is swept. Given κ , the response coordinates are identified only up to a rotation: the isotropic prior of Equation (3) and the linear decoder are invariant under $\mathbf{w} \mapsto \mathbf{Q}\mathbf{w}$, $\mathbf{B} \mapsto \mathbf{B}\mathbf{Q}^\top$, $m_\psi \mapsto \mathbf{Q}m_\psi$ for orthogonal $\mathbf{Q}$ (the learned prior mean absorbs the last), while the fixed prior scale removes the rescaling freedom

(Section 2.3); the diagonal posterior of Equation (14) is not rotation-equivariant, so this is a symmetry of the model, not of the fitted objective. Through the softmax, each column of $\mathbf{B}$ is further identified only up to an additive constant over genes, and so is $a(\mathbf{z})$ per cell; loadings are read only after centring each column over genes. The allocation between $\mathbf{z}$ and $\mathbf{w}$ is not pinned by the likelihood, since both posteriors see the counts; it rests on the conditional-invariance constraint, on the intrinsic path, and on $d_w < d_z$. For $\mathbf{w}$ given the context this is in the spirit of the auxiliary-variable setting of Khemakhem et al. (2020), whose guarantee (recovery up to an affine map, under an injective additive-noise decoder and sufficient variability of the conditional prior) we do not establish for this likelihood; for $\mathbf{z}$ no comparable guarantee exists without the constraint (Locatello et al., 2019), and the constraint is what we verify with the probe rather than assume.

The sweep. The unit of inference is a sensitivity analysis with κ as the sensitivity parameter, in the spirit of partial identification (Manski, 2003; Rosenbaum, 2002). The model is fitted independently at each $\kappa \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$ with several random seeds per grid point (three in the runs reported here, Section 3.4), with every other weight held at the values calibrated at $\kappa = 0.1$ (Appendix E) and the held-out probe checked at every κ so that a change in a readout is not a change in the invariance. Every readout of interest, the loadings $\mathbf{B}$, the per-type magnitude of the response, the spatial autocorrelation of each latent, the held-out probes and the agreement with external labels, is reported as a trajectory over κ with its envelope over seeds. We call a readout stable when its sign and, for rankings, its order are unchanged over the grid and its seed envelope excludes zero at every grid point [TODO: confirm this rule]. A stable readout is a finding; one that changes sign, vanishes, or is inconsistent across seeds is not separable from leakage on this data. Some readouts are expected to move with κ by construction: as κ grows, more of the resemblance between neighbours is assigned to the mixture and the fitted response magnitude shrinks, so a decline in $\|\mathbf{w}\|$ along the grid is mechanical, whereas a change in the *direction* of a programme or in which latent a target aligns with is not. Two consequences follow. Gene programmes are identified only up to the rotation and the per-column constants above, so raw loadings and probe directions are never compared across runs or seeds; only run-internal, rotation-invariant summaries are reported, and any gene-level statement is treated as relative to the run that produced it. And the reported range is not a confidence interval: it does not shrink with more cells, because the confound it expresses is not a sampling problem. An operating point, $\kappa = 0.1$ here,

is used for figures and for the analyses that require a single fit, and is always reported alongside the sweep.

What $\mathbf{w}$ is not. The context $\mathbf{c}_i$ is the environment: deterministic, shared by co-located cells, a conditioning variable. If a niche descriptor were all that was wanted, one would drop the latent and feed $\mathbf{c}_i$ to the decoder directly, which is roughly the node-centric model of Fischer et al. (2023) and is simpler. The response $\mathbf{w}_i$ is latent and per cell, and it earns its place only through within-niche heterogeneity: anomalies, per-cell responses, and counterfactuals with abduction. Whether it does so at a given operating point is an empirical question that the per-dimension divergence answers, and we report it as such (Section 3.6).

3 EXPERIMENTS

[TODO: This section is a structured skeleton. Every experiment below is described by its goal and its protocol; numbers are deliberately withheld until the multi-seed reruns are complete. The existing runs (one section, six leak fractions, three seeds, one long reference fit) are the starting point and inform the protocol text, not the reported values.]

3.1 Data and Preprocessing

Sections. The primary section is a publicly available formalin-fixed paraffin-embedded ovarian carcinoma section profiled on the Xenium platform [NEEDS A SOURCE] with a 5,001-gene panel plus 100 custom targets: 407,120 segmented cells after removing four without a usable boundary polygon, a median of 178 transcripts and 147 detected genes per cell (Appendix D.1). A second section [PENDING RUNS: lung carcinoma section, same panel family, unsupervised cluster labels] is used to test whether the decomposition transfers to a tissue with no curated annotation.

Labels. On the ovarian section t_i is the vendor-supplied curated annotation with $K = 18$ classes, one of which is an “unassigned” class that we keep as a type in the model but exclude from all biological readouts. As a label-set control we refit the ovarian section with an unsupervised graph-based clustering ($K = 26$) in place of the curated annotation (Appendix H.1) [PENDING RUNS: multi-seed].

Graph, kernel, image embedding. Delaunay adjacency with Voronoi faces, pruned at $40 \mu\text{m}$; kernel β with $\tau = 20 \mu\text{m}$; image embedding from a pathology foundation model for multiplexed immunofluorescence

(Shaban et al., 2025) applied to a $128\mu\text{m}$ patch at $0.5\mu\text{m}$ per pixel in which a $25\mu\text{m}$ disc around the cell centroid is masked. Every choice in this pipeline, including the measurement that a contact-based graph is degenerate on this platform, the selection of the mask radius from the distribution of cell covering radii, and the test that the masked embedding carries no cell-type information beyond neighbour homophily, is reported in Appendix D.

Splits. Cells are partitioned into contiguous rectangular tiles of at most 4,096 cells by recursive median splits; 15% of tiles are held out and serve every held-out readout and the stopping rule. There is no third split. **[TODO: decide whether an untouched test set of tiles is required for the final claims]**

3.2 Operating Point and Calibration

Goal. Fix every weight in Equation (17) other than κ on the primary section before the sweep, by the probe protocol of Section 2.5, so that the sweep varies one thing.

Protocol. Short fits at $\kappa = 0.1$ scanning α_a under the closed-form penalty, then α_w and ω ; a convergence-length check of the closed-form operating point with both probe families; escalation to the adversary and a scan of its strength and update schedule; the operating point chosen at the crossing of held-out ΔCE and $\mathbf{z}$ -type agreement. The full calibration history, including the two knife-edges in α_w and α_a and the observation that an under-trained adversary is a placebo, is in Appendix E.

Result. **[PENDING RUNS: restate the operating point with seed envelopes once the calibration scans are repeated with three seeds]**

3.3 Synthetic Recovery

Goal. Show that the model recovers planted intrinsic states, responses and gene programmes from data simulated from its own generative story, with and without leakage, and that the invariance penalty keeps the planted niche out of $\mathbf{z}$.

Protocol. See Appendix F. **[PENDING RUNS: multi-seed recovery at planted $\kappa \in \{0, 0.2\}$; recovery as a function of the assumed κ when it is misspecified]**

3.4 The Leakage Sweep

Goal. Report every readout as a trajectory over κ with seed envelopes, and identify which survive.

Readouts. Held-out reconstruction (globally and stratified by the number of edges a cell lost to pruning); the within-type mirror coefficient against its permuted control; $\mathbf{z}$ -type agreement; held-out probe ΔCE in both families against the noise floor; the stability of $\mathbf{B}$ within and across seeds after optimal column matching; the per-type magnitude of the response; per-dimension spatial autocorrelation of both latents; the held-out niche probes of Section 3.5; details in Appendices H.2 to H.4.

Result. **[PENDING RUNS: leakage-sweep re-run with at least five seeds per κ ; second section]**

3.5 What $\mathbf{z}$ Carries and What $\mathbf{w}$ Carries

Goal. Every analysis here is a target variable with a known allegiance, probed identically from both latents against reference rows: a within-type permutation floor, a depth-only baseline (the probe from $\log \ell$ alone, since depth varies across tissue), and, for expression-derived targets, a linear reference from the leading principal components of the counts. The claim is always the asymmetry between the two latents, never one number. Spatial-block cross-validation is used throughout.

Intrinsic target: cell-cycle phase. Continuous S and G2/M scores from standard gene sets (Tirosh et al., 2016), regressed within type from μ_z , μ_w and the references on the types with the highest fraction of proliferating cells (Appendix G). **[PENDING RUNS: multi-seed numbers]**

Spatial targets. (i) A discrete niche label from k -means on neighbour composition (10 niches), predicted per type by class-balanced multinomial regression from each latent; (ii) per-dimension Moran’s I (Moran, 1950) of each latent after per-type centring, against a permutation null, with collapsed response dimensions flagged; (iii) a principal-curve pseudotime in each latent space and its agreement with the tissue gradient after partialling out type; (iv) distance to anatomical landmarks (vasculature, compact smooth-muscle structures, the tumour–stroma interface) in a mid-range band where one-hop context cannot see the landmark, with an additional baseline from raw neighbour composition because the landmark sets are themselves type-defined (Appendix G). **[PENDING RUNS: multi-seed numbers]**

Headline figure. **[TODO: figure to be produced from the reruns]** An allegiance matrix with the targets as rows and the two latents as columns, each cell certified by its references, expected to be block-diagonal; a companion plot of the key numbers against κ with seed envelopes.

3.6 What the Response Looks Like at the Operating Point

Goal. Report honestly how much per-cell information $\mathbf{w}$ carries beyond its prior at the operating point: the coefficient of determination of μ_w on $m_\psi(\mathbf{c}, t)$, the per-dimension divergence, and the effective dimensionality of the response (Appendix H.5). **[PENDING RUNS: multi-seed numbers]**

4 CONCLUSION

[TODO: Deliberately empty for now.]

Future work. **[TODO: Perturbations as a third source of variation: with a perturbed and an unperturbed condition the model gains a third variable, and the decomposition becomes intrinsic state, microenvironmental response and perturbation response, with the leakage sweep applied to all three. Per-cell anomaly scores and counterfactuals with abduction once an operating point with a non-degenerate response posterior is established. Estimation of κ from a nuclear/extranuclear split of the counts.]**

References

- Bengio, Y., Léonard, N., and Courville, A. (2013). Estimating or propagating gradients through stochastic neurons for conditional computation. *arXiv preprint arXiv:1308.3432*.
- Birk, S., Lotfollahi, M., et al. (2025). Quantitative characterization of cell niches in spatially resolved omics data. *Nature Genetics*. CHECK: author list, volume, identifier.
- Brody, S., Alon, U., and Yahav, E. (2022). How attentive are graph attention networks? In *International Conference on Learning Representations*.
- Cover, T. M. and Thomas, J. A. (2006). *Elements of Information Theory*. Wiley, 2nd edition.
- Dong, M., Kluger, Y., et al. (2025). SIMVI disentangles intrinsic and spatial-induced cellular states in spatial omics data. *Nature Communications*. CHECK: author list, volume, identifier.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houshy, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*.
- Elazar, Y. and Goldberg, Y. (2018). Adversarial removal of demographic attributes from text data. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 11–21.
- Ergen, C., Yosef, N., et al. (2025). resolVI: addressing noise and bias in spatial transcriptomics. *bioRxiv*. CHECK: exact title, author list, and identifier.
- Ester, M., Kriegel, H.-P., Sander, J., and Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. In *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining*, pages 226–231.
- Fischer, D. S., Schaar, A. C., and Theis, F. J. (2023). Modeling intercellular communication in tissues using spatial graphs of cells. *Nature Biotechnology*, 41:332–336.
- Ganchev, K., Graça, J., Gillenwater, J., and Taskar, B. (2010). Posterior regularization for structured latent variable models. *Journal of Machine Learning Research*, 11:2001–2049.
- Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., and Lempitsky, V. (2016). Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35.
- Hand, D. J. and Till, R. J. (2001). A simple generalisation of the area under the ROC curve for multiple class classification problems. *Machine Learning*, 45:171–186.
- Hastie, T. and Stuetzle, W. (1989). Principal curves. *Journal of the American Statistical Association*, 84(406):502–516.
- Janesick, A., Shelansky, R., Gottscho, A. D., Wagner, F., Williams, S. R., Rouault, M., Beliakoff, G., Morrison, C. A., Oliveira, M. F., Sicherman, J. T., et al. (2023). High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and in situ analysis. *Nature Communications*, 14:8353. CHECK: full author list truncated.
- Khemakhem, I., Kingma, D. P., Monti, R. P., and Hyvärinen, A. (2020). Variational autoencoders and nonlinear ICA: A unifying framework. In *International Conference on Artificial Intelligence and Statistics*.
- Kingma, D. P. and Ba, J. (2015). Adam: A method for stochastic optimization. In *International Conference on Learning Representations*.
- Kingma, D. P., Salimans, T., Jozefowicz, R., Chen, X., Sutskever, I., and Welling, M. (2016). Improved variational inference with inverse autoregressive flow. In *Advances in Neural Information Processing Systems*, volume 29.

- Kingma, D. P. and Welling, M. (2014). Auto-encoding variational Bayes. In *International Conference on Learning Representations*.
- Klys, J., Snell, J., and Zemel, R. (2018). Learning latent subspaces in variational autoencoders. In *Advances in Neural Information Processing Systems*, volume 31.
- Kuhn, H. W. (1955). The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97.
- Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.
- Locatello, F., Bauer, S., Lucic, M., Rätsch, G., Gelly, S., Schölkopf, B., and Bachem, O. (2019). Challenging common assumptions in the unsupervised learning of disentangled representations. In *International Conference on Machine Learning*.
- Loshchilov, I. and Hutter, F. (2017). SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations*.
- Louizos, C., Swersky, K., Li, Y., Welling, M., and Zemel, R. (2016). The variational fair autoencoder. In *International Conference on Learning Representations*.
- Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer.
- Moran, P. A. P. (1950). Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2):17–23.
- Rezende, D. J., Mohamed, S., and Wierstra, D. (2014). Stochastic backpropagation and approximate inference in deep generative models. In *International Conference on Machine Learning*.
- Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillerá-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., and Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929.
- Rosenbaum, P. R. (2002). *Observational Studies*. Springer, 2nd edition.
- Schäfer, J. and Strimmer, K. (2005). A shrinkage approach to large-scale covariance matrix estimation and implications for functional genomics. *Statistical Applications in Genetics and Molecular Biology*, 4(1).
- Shaban, M. et al. (2025). KRONOS: A foundation model for spatial proteomics. *arXiv preprint*. CHECK: exact title, authors, and identifier; two model generations were used.
- Slavutsky, Y., Beker, O., Blei, D., and Dumitrescu, B. (2026). Variational learning of disentangled representations. In *International Conference on Machine Learning*. arXiv:2506.17182.
- Strehl, A. and Ghosh, J. (2002). Cluster ensembles — a knowledge reuse framework for combining multiple partitions. *Journal of Machine Learning Research*, 3:583–617.
- Tirosh, I., Izar, B., Prakadan, S. M., Wadsworth, M. H., Treacy, D., Trombetta, J. J., Rotem, A., Rodman, C., Lian, C., Murphy, G., Fallahi-Sichani, M., Dutton-Regester, K., Lin, J.-R., Cohen, O., Shah, P., Lu, D., Genshaft, A. S., Hughes, T. K., Ziegler, C. G. K., Kazer, S. W., Gaillard, A., Kolb, K. E., Villani, A.-C., Johannessen, C. M., Andreev, A. Y., Van Allen, E. M., Bertagnolli, M., Sorger, P. K., Sullivan, R. J., Flaherty, K. T., Frederick, D. T., Jané-Valbuena, J., Yoon, C. H., Rozenblatt-Rosen, O., Shalek, A. K., Regev, A., and Garraway, L. A. (2016). Dissecting the multicellular ecosystem of metastatic melanoma by single-cell RNA-seq. *Science*, 352(6282):189–196.
- Tomczak, J. M. and Welling, M. (2018). VAE with a VampPrior. In *International Conference on Artificial Intelligence and Statistics*.
- Xie, Q., Dai, Z., Du, Y., Hovy, E., and Neubig, G. (2017). Controllable invariance through adversarial feature learning. In *Advances in Neural Information Processing Systems*, volume 30.

Checklist

[TODO: Answer at submission time. Questions must not be modified; the instruction paragraph of the template has been removed.]

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes]
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes]
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. **[TODO: answer]**
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes]
 - (b) Complete proofs of all theoretical results. [Yes]

- (c) Clear explanations of any assumptions. [Yes]
3. For all figures and tables that present empirical results, check if you include:
- (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). **[TODO: answer]**
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes]
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). **[TODO: answer]**
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
- (a) Citations of the creator If your work uses existing assets. [Yes]
 - (b) The license information of the assets, if applicable. **[TODO: answer]**
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Not Applicable]
 - (d) Information about consent from data providers/curators. [Not Applicable]
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
- (a) The full text of instructions given to participants and screenshots. [Not Applicable]
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

DISCELL: Supplementary Materials

A DERIVATIONS

A.1 The Reconstruction Bound

Fix a cell i and hold $\mathbf{c}_i$ and $\bar{\rho}_i$ constant, as Section 2.6 does. Then

$$\log p(\mathbf{x}_i | \mathbf{c}_i, t_i) = \log \iint p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) p(\mathbf{w} | \mathbf{c}_i, t_i) p(\mathbf{z}) d\mathbf{z} d\mathbf{w} \quad (22)$$

$$\geq \mathbb{E}_{q(\mathbf{z})} \mathbb{E}_{q(\mathbf{w}|\mathbf{z})} \left[\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) + \log p(\mathbf{w} | \mathbf{c}_i, t_i) + \log p(\mathbf{z}) - \log q(\mathbf{z}) - \log q(\mathbf{w} | \mathbf{z}) \right] \quad (23)$$

$$= \mathbb{E}_{q(\mathbf{z})} \left[\mathbb{E}_{q(\mathbf{w}|\mathbf{z})} \left[\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) \right] - \text{KL}(q(\mathbf{w} | \mathbf{z}) \| p(\mathbf{w} | \mathbf{c}_i, t_i)) \right] - \text{KL}(q(\mathbf{z}) \| p(\mathbf{z})), \quad (24)$$

where Equation (23) is Jensen’s inequality under the variational family $q(\mathbf{z}) q(\mathbf{w} | \mathbf{c}_i, t_i, \mathbf{z}, \mathbf{x}_i)$. Because $q(\mathbf{w} | \cdot)$ conditions on the sampled $\mathbf{z}$, the exact bound keeps the $\mathbf{w}$ -divergence inside $\mathbb{E}_{q(\mathbf{z})}$. The divergence between two diagonal Gaussians is available in closed form,

$$\text{KL}(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2) \| \mathcal{N}(\mathbf{m}, \mathbf{s}^2)) = \sum_k \left[\log \frac{s_k}{\sigma_k} + \frac{\sigma_k^2 + (\mu_k - m_k)^2}{2s_k^2} - \frac{1}{2} \right], \quad (25)$$

and evaluating it at the single reparameterised draw of $\mathbf{z}$ used for the reconstruction term is an unbiased one-sample estimate of $\mathbb{E}_{q(\mathbf{z})}[\text{KL}(\cdot)]$.

A.2 The Intrinsic Path and the Factor $(1 + \omega)$

A second bound on the same data holds for the model in which $\mathbf{w}$ is not inferred. Setting $q(\mathbf{w}) := p(\mathbf{w} | \mathbf{c}_i, t_i)$ zeroes the $\mathbf{w}$ -divergence and leaves

$$\log p(\mathbf{x}_i | \mathbf{c}_i, t_i) \geq \mathbb{E}_{q(\mathbf{z})} \mathbb{E}_{p(\mathbf{w}|\mathbf{c}_i, t_i)} \left[\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) \right] - \text{KL}(q(\mathbf{z}) \| p(\mathbf{z})) =: L_i^z. \quad (26)$$

Call the bound of Appendix A.1 L_i . Each carries its own $-\text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i))$, so $L_i + \omega L_i^z$ contains $-(1 + \omega) \text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i))$. Omitting the second copy breaks the correspondence with $L_i + \omega L_i^z$ even at $\omega = 1$.

The construction mirrors the two-bound objective of Slavutsky et al. (2026), where the sum of the shared-only and joint bounds corresponds to minimising $\text{KL}(q(\mathbf{z}) \| p(\mathbf{z} | \mathbf{x})) + \text{KL}(q(\mathbf{z})q(\mathbf{w}) \| p(\mathbf{z}, \mathbf{w} | \mathbf{x}, u))$ with u their condition label, a sum of two non-negative divergences. It is coherent as an M-estimation target, but it is not the evidence lower bound of a single model, so q is not the posterior of anything, which is one of the reasons uncertainty is reported from the leakage sweep rather than from q .

Why the intrinsic path is load-bearing: in Slavutsky et al. (2026) a structured prior couples the two codes through the condition and thereby rewards the shared code for carrying signal [TODO: state the exact form of that prior]. Our prior on $\mathbf{w}$ never references $\mathbf{z}$, and the prior on $\mathbf{z}$ is a standard normal, so nothing else in the objective rewards $\mathbf{z}$ for being informative. The intrinsic path supplies that pressure directly, since under it $\mathbf{z}$ must explain the cell given only the expected response. It runs through the same leakage mixture as the first term, so it never asserts $\kappa = 0$ and contradicts it.

A.3 The Invariance Term as Posterior Regularisation

Neither Pen nor Adv $_i$ is part of any bound. Both are posterior regularisation in the sense of Ganchev et al. (2010): a constraint on the variational family, $\text{I}(\mathbf{z}; [\mathbf{y}, \Phi] | t) = 0$, written as a penalty on the variational objective. This is the same status Slavutsky et al. (2026) give their adversarial term. It is a statement about which q are admissible, not a modification of the generative model, and the generative model’s conditional independence $\mathbf{z}_i \perp [\mathbf{y}_i, \Phi_i] | t_i$ is what it encodes.

A.4 The Gaussian Conditional Mutual Information

For jointly Gaussian $(\mathbf{z}, \mathbf{v})$ with covariance $\Sigma_{[\mathbf{z}, \mathbf{v}]}$ and marginal covariances $\Sigma_{\mathbf{z}}, \Sigma_{\mathbf{v}}$, the mutual information is $I(\mathbf{z}; \mathbf{v}) = \frac{1}{2}[\log \det \Sigma_{\mathbf{z}} + \log \det \Sigma_{\mathbf{v}} - \log \det \Sigma_{[\mathbf{z}, \mathbf{v}]}]$ (Cover and Thomas, 2006). Equation (19) applies this per type and averages with the type frequencies, which is the Gaussian estimate of $I(\mathbf{z}; \mathbf{v} | t)$ under a type-wise Gaussian assumption. Two properties used in Section 2.5 follow directly. First, the value is invariant to a per-dimension rescaling $\mathbf{u} \mapsto \mathbf{D}\mathbf{u}$ with $\mathbf{D}$ diagonal: each log-determinant gains $\log \det$ of the corresponding block of $\mathbf{D}^2$, and the blocks of $\mathbf{D}$ for $\mathbf{z}$ and $\mathbf{v}$ together make up $\mathbf{D}$, so the three terms cancel exactly. The penalty may therefore be computed on correlation matrices. Second, on the simplex $\sum_k y_k = 1$ the covariance of the full composition vector is singular and the log-determinant is $-\infty$; dropping one column removes the constraint without changing the information, since the dropped column is a deterministic function of the others.

A.5 Feedback Amplification Through the Leakage Path (Heuristic)

The following is a first-order heuristic, not a theorem. Write the decoder output of cell j as ρ_j and suppose gradients were allowed to flow from cell i 's reconstruction through $\bar{\rho}_i = \sum_j \beta_{ij} \rho_j$ into the parameters that produce ρ_j . Linearising around a fixed point of the coupled system, a perturbation δ of a cell's clean composition induces a perturbation $\kappa \beta \delta$ of its neighbours' fitted compositions, which feeds back through the same path; the total response is $\delta \sum_{n \geq 0} (\kappa \mathbf{P}_\beta)^n$ for a row-stochastic $\mathbf{P}_\beta$ built from β , whose spectral radius is at most one, so the geometric series is bounded by $1/(1 - \kappa)$. The stop-gradient truncates the series at $n = 0$. The argument concerns perturbations of the fitted compositions; the gradient fixed point through shared decoder parameters is more involved, and the empirical fact that matters is the collapse reported by Ergen et al. (2025) when the path is left open. **[TODO: either complete the gradient-level argument or keep this as a heuristic]**

B DESIGN RATIONALE

This appendix carries the parts of the arguments of Section 2 that the main text only summarises. It does not repeat what is already there.

B.1 The Mirror Attractor

The attention layer of Equation (10) computes logits $e_{ij} = \mathbf{a}^\top \text{LeakyReLU}(\mathbf{W}_{\text{dst}} \mathbf{h}_i + \mathbf{W}_{\text{src}} \mathbf{h}_j)$ and aggregates $\sum_j \pi_{ij} \mathbf{W}_{\text{src}} \mathbf{h}_j$, so the destination feature $\mathbf{h}_i$ enters the *attention* but never the *value* (Brody et al., 2022). With $\mathbf{z}_i$ as the destination feature, the selection channel of Section 2.3 opens. What the type query gives up is state-specific attention, which belongs in the decoder (as type-dependent loadings or a low-rank interaction with $\mathbf{z}$) rather than in the context encoder.

The diagnostic is the within-type coefficient of determination of $\mathbf{z}_i$ regressed on $\mathbf{c}_i$ against a within-type permuted control (Section 2.7), or the correlation between π_{ij} and $\text{sim}(\mathbf{z}_i, \mathbf{z}_j)$. Downstream, a mirror is indistinguishable from ordinary posterior collapse of $\mathbf{w}$; a free-bits remedy (Kingma et al., 2016) would mask it and leave the anomaly score meaningless, which is why the mirror is ruled out first.

Zeroing $\mathbf{z}_i$ while keeping self-loops is not an alternative: the softmax still spends attention mass on the self-edge, down-weighting neighbours, and under $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ the origin is the prior mean, so a zero vector asserts “this cell is average” rather than “ignore this cell”.

B.2 Why the Context Gets No Per-Edge Geometry, and Why the Graph Is Delaunay

A discretised contact indicator (touching, near, far) looks as if it dodges the collinearity objection of Section 2.3, but on this platform under nuclear-expansion segmentation whether two segmented polygons touch is largely a measurement of local density: dense regions touch, sparse regions do not. The indicator would smuggle density back in, in the confounded direction, and a tolerance parameter only moves the threshold. The measurements that establish this on our data are in Appendix D.2.

Delaunay adjacency resolves it: two cells are neighbours if and only if their Voronoi cells share a face, which is the correct notion of adjacency for a packing, built from centroids alone. The one parameter that returns is the pruning of long edges, and it is much less sensitive than choosing a neighbour count. The Voronoi face length

is used in the kernel rather than shared segmented boundary because the Voronoi face is where the boundary between two cells effectively lies, its length is a contact-extent proxy defined whether or not the polygons touch, and it is independent of the expansion setting.

The fallback is empirical. If m_ψ predicts responses poorly, or held-out reconstruction is worse than a comparison run whose attention layer carries edge features (distance, face length), that is direct evidence that the continuous signal mattered. If it is added back, the correlation between π_{ij} and β_{ij} must be monitored: a high correlation means the attention has rediscovered the leakage kernel and $\mathbf{w}$ is becoming collinear with $\bar{\rho}$. That risk is tolerable while κ is fixed, since there is no free contamination parameter to feed, and intolerable once κ is estimated. **[PENDING RUNS: the edge-feature comparison run has not been performed]**

B.3 Three Routes to a Vanishing Response Divergence

Cause	Check	Fix
Posterior collapse	α_w large relative to reconstruction	lower α_w ; free bits
Mirror	within-type R^2 of $\mathbf{z}_i$ on $\mathbf{c}_i$ vs. permuted control	type embedding as query
Starved of ego evidence	does the response posterior receive $\mathbf{x}_i$ directly?	add the direct path

B.4 Conditional Invariance and Label Granularity

The marginal-invariance failure is documented by Slavutsky et al. (2026) for the conditional subspace VAE (Klys et al., 2018), whose invariance drives the agreement between the shared code and the nuisance variable to zero while collapsing its agreement with cell type **[TODO: quote the exact numbers from that comparison]**. Coarse labels are preferable to fine ones for three reasons: fine labels leave $\mathbf{z}$ with nothing to carry, they block more of the position→fate→expression path, and they destroy the overlap that any conditional comparison needs. The label-set control of Appendix H.1 varies the granularity.

B.5 The Leak Fraction: Grid Range and Upgrade Path

The grid range comes from three sources: zero as anchor; published misassignment estimates for this platform **[NEEDS A SOURCE]**; and a marker-set ceiling. For an atlas-defined set $\mathcal{M}$ of genes that a cell of type t_i should not express, and an empirical neighbour profile $\tilde{\rho}_i$ obtained by averaging the observed compositions of i 's neighbours with the kernel β , the ratio $\sum_{g \in \mathcal{M}} x_{ig} / (\ell_i \sum_{g \in \mathcal{M}} \tilde{\rho}_{ig})$ bounds the leak fraction from above. The set must be defined externally: ranking genes by their neighbour-to-own expression ratio from the same counts selects the zero-count genes and drives the bound to zero with probability one. **[PENDING RUNS: compute the marker-set ceiling on both sections]**

This version of the model sweeps κ ; it cannot estimate it. Estimation needs two views of one cell that share a foreign profile at different contamination levels, which a nuclear/extranuclear split of the transcript coordinates provides. Its identifying condition is that gene-wise nuclear retention varies across genes; with constant retention both views lie on the same line between ρ_i and $\bar{\rho}_i$ and nothing is pinned. Adding it does not restructure the model: the likelihood becomes a multinomial over $2G$ bins and κ moves from hyperparameter to latent. We leave it as future work.

B.6 Learned Variances: What to Learn and What Not To

If a conditional $\sigma_w(\mathbf{c}, t)$ is wanted, because it would standardise deviation against local variability so that a deviation of two in a niche where everyone deviates by two does not read as anomalous, it must be bounded, e.g. $\sigma_w \in [0.5, 2]$ through a scaled sigmoid, so that it cannot inflate away, and its mean over cells must be checked for upward drift over training.

The prior on $\mathbf{z}$ is a standard normal. A richer prior, a mixture with one component per type or a VampPrior with archetypal pseudo-inputs (Tomczak and Welling, 2018), would be a genuine upgrade for cell-type structure and would make t a latent rather than an input, but that is a different change, not a variance question.

C IMPLEMENTATION DETAILS

C.1 Architecture and Hyperparameters

Table 2 lists every network and every fixed quantity as used in the experiments. All multilayer perceptrons use rectified-linear hidden units.

C.2 Choices Made in Implementation

Where the model description of Section 2 leaves a choice open, Table 3 records what was done and why. Every entry is also stated in the method text; the table collects them in one place. **[TODO: author to decide whether this table stays; it duplicates the method text]**

D DATA AND PREPROCESSING

[TODO: Numbers in this appendix are from the existing single-section preprocessing and are stable (they do not depend on model seeds). They will be duplicated for the second section.]

D.1 Sections and Labels

The ovarian section: 407,124 segmented cells in the vendor output, of which four lack a usable boundary polygon and are dropped; 5,101 features, all treated as genes in the multinomial (a 5,001-gene predesigned panel plus 100 custom targets); median 178 transcripts and 147 detected genes per cell (mean 261.6 transcripts; 5th and 95th percentiles 18 and 761); median cell area $57 \mu\text{m}^2$, median equivalent diameter $8.5 \mu\text{m}$. Segmentation provenance: 69% of cells from an interior stain, 29% from a boundary stain, 2% from nuclear expansion. Labels: the vendor’s curated annotation covers 406,611 cells in 18 classes including an “unassigned” class; the 513 cells absent from the annotation are folded onto that class. Type sizes range from 103,606 (tumour cells) to 1,961 (urothelial-like). An unsupervised graph-based clustering with 26 clusters, supplied with the dataset, serves as the label-set control.

[PENDING RUNS: lung section: cell count, panel, labels from unsupervised clustering]

D.2 Why a Contact Graph Is Degenerate on This Platform

On a 20,000-cell subset of a lung section, requiring exact polygon contact yields 1,448 edges, a mean degree of 0.14 and 87.7% isolated cells; a $1 \mu\text{m}$ tolerance yields 25,956 edges, mean degree 2.6 and 10.8% isolated; the Voronoi graph yields 58,206 edges, mean degree 5.8 and 0.1% isolated. A quarter of neighbouring polygon pairs sit within $0.12 \mu\text{m}$ of one another but only 2.5% touch at zero distance. On the full ovarian section the $1 \mu\text{m}$ -tolerance contact graph has 634,557 edges, mean degree 3.1, 9.6% isolated cells and a median shared-boundary length of zero, so a leakage kernel built from shared segmented boundary would be all-zero for 94.5% of connected cells; the Voronoi graph has 1,197,730 edges, mean degree 5.9, 0.13% isolated cells and a median face length of $6.85 \mu\text{m}$. The Voronoi face length correlates only weakly with the polygon gap ($r = -0.11$), which is the property the kernel needs: contact extent that is not a density readout.

D.3 Pruning and the Kernel

The 95th percentile of Voronoi edge length on the ovarian section is $27.4 \mu\text{m}$. Pruning at $40 \mu\text{m}$ removes 1.11% of edges (13,281 of 1,197,730), isolates 372 additional cells ($527 \rightarrow 899$), leaves 1,511 cells with a single neighbour, and costs no cell more than ten edges. The composition $\mathbf{y}$ and the type table $\bar{\mathbf{y}}(t)$ are recomputed on the pruned graph. The decay length τ is inert at this cell spacing: the perplexity of the kernel rows (effective number of contributing neighbours) is 4.39, 4.68 and 4.90 at $\tau = 10 \mu\text{m}$, $20 \mu\text{m}$ and no decay, against a median degree of 6; β is dominated by the face lengths, and conclusions from the leakage sweep should not be attributed to τ . The 2,410 cells with at most two neighbours after pruning receive the full leak fraction against a one- or two-neighbour influx; they are excluded from effect readouts, and their extra reconstruction cost along the sweep is reported separately (Appendix H.2).

D.4 Image Embedding

The morphology image has four channels (nuclear stain, a boundary-stain cocktail, an interior ribosomal-RNA stain, and a smooth-muscle/mesenchymal cocktail). Patches of 256 pixels at $0.5\ \mu\text{m}$ per pixel (a $128\ \mu\text{m}$ field) are embedded with a marker-aware vision transformer (Dosovitskiy et al., 2021) pretrained on multiplexed immunofluorescence (Shaban et al., 2025); two generations of that model score identically on a class-separation test ($+0.418$ versus $+0.417$ mean within-minus-between class cosine, 36.2% nearest-neighbour label agreement for both, chance $1/18$), so the faster 384-dimensional one is used. Scaling the 16-bit intensities to $[0, 1]$ before embedding mattered more than the choice of model (separation $+0.272 \rightarrow +0.418$). The channel-to-marker mapping is approximate for two of the four channels and the ribosomal stain is outside the model’s marker vocabulary; this is a known limitation. [TODO: state as limitation in main text?]

Mask radius. The ego cell is hidden by zeroing a fixed disc around its centroid. A disc rather than the polygon, so that the hole carries no silhouette; zeros rather than inpainting. The radius is chosen from the distribution of covering radii (the smallest centroid-centred disc containing the cell) over all 407,120 cells: median $6.3\ \mu\text{m}$, 99th percentile $14.3\ \mu\text{m}$, 99.99th percentile $22.6\ \mu\text{m}$, maximum $36.8\ \mu\text{m}$. A $15\ \mu\text{m}$ disc leaves 2,740 cells partially visible; $25\ \mu\text{m}$ covers all but six, which are excluded from the embedding and receive a zero image vector. The disc is 12% of the patch area and, at this cell density, removes a median of 18 neighbouring cells, so the masked embedding describes tissue beyond roughly $25\ \mu\text{m}$ rather than “the patch minus the cell”. [TODO: state this scale explicitly wherever Φ is interpreted]

D.5 The Class Test for the Image Context

Goal. Establish what cell-type information the masked image embedding carries, and in particular whether it reads the ego cell through the hole or through its silhouette. The intended reading of Φ_i is a description of the surroundings; if a masked patch predicted the ego cell’s type better than its neighbours’ labels do, the embedding would be a second copy of the cell and not a context.

Protocol. Seven embeddings are computed on identical patches: unmasked; ego-masked ($25\ \mu\text{m}$ disc zeroed); ego-only (everything outside the cell’s polygon zeroed); each for both model generations; plus ego-only at native resolution ($0.2125\ \mu\text{m}$ per pixel). Each embedding is scored by a multinomial logistic regression for the 18 types (standardised features, unit regularisation), reported as macro one-versus-rest area under the ROC curve (Hand and Till, 2001), under a spatial split into $1,000\ \mu\text{m}$ tiles with 30% of tiles held out and cells within half a patch of a tile edge dropped (92 tiles, 23 held out, 95,506 edge cells dropped, 40,000 training cells, 79,741 held-out cells). The reference is the same classifier from neighbour composition $\mathbf{y}_i$ alone, i.e. type homophily.

Result. Homophily alone reaches 0.908. The unmasked patch reaches 0.850 and the ego-masked patch 0.829 (first model generation; 0.849 and 0.818 for the second): masking costs 0.02, and both context arms sit roughly 0.08 below the homophily reference, so the masked context carries no type information beyond what the neighbour labels already carry. The cell’s own image reaches 0.873 (0.893 at native resolution; 0.896 for the second generation), i.e. the cell alone beats the whole patch. This is the intended, legitimate regime: a context that sits below the homophily baseline is exogenous to the cell, and its value to the model is not type information but the architectural signal that neighbour labels lack, which the reconstruction ablation of Appendix E measures. [PENDING RUNS: repeat the classifier with three seeds of the split for error bars; second section]

E CALIBRATION OF THE OPERATING POINT

[TODO: This appendix currently records the calibration history from single-seed short fits. Every scan will be repeated with three seeds before numbers are quoted in the main text.]

Protocol. All scans are on the ovarian section at $\kappa = 0.1$, with the architecture of Table 2, 40-epoch fits and one evaluation, unless stated. Two held-out probes are fitted to each model (Equation (21)): a ridge regression and a two-hidden-layer perceptron of width 64, both from $[\mu_z, \text{onehot}(t)]$ to the invariance block, both trained on training tiles and evaluated on held-out tiles. The uncontrolled baseline is the model at $\alpha_a = 0$; the noise floor is the probe with μ_z permuted within type. The $\mathbf{z}$ -type agreement is the normalised mutual information (Strehl and Ghosh, 2002) between a k -means clustering of μ_z ($k = K$) and the labels.

Round 1: closed-form penalty strength. $\alpha_a \in \{0, 0.01, 0.02, 0.05, 0.1\}$. The nonlinear probe reads roughly 2.5 to 3 times the linear one at every setting, so all decisions are taken on the nonlinear probe. The agreement with type collapses between $\alpha_a = 0.03$ and 0.04 (a cliff from 0.56 to 0.35) while the residual leak falls from 28% to 6% of the uncontrolled baseline; 0.03 is the closed-form operating point. [PENDING RUNS: three seeds]

Image ablation. At the closed-form operating point, replacing Φ with zeros costs 0.0092 nats per count of held-out reconstruction (about 1.6 nats per cell) and the type agreement falls from 0.56 to 0.35. This is the evidence for keeping the full-dimensional embedding in $\mathbf{c}_i$ rather than a low-dimensional projection. [PENDING RUNS: three seeds; a projected-embedding arm at the operating point]

Round 2: α_w and ω . α_w is a knife-edge. At 0.03 the direct path from the counts lets $\mathbf{w}$ steal identity (agreement 0.33, per-dimension divergence 0.044); at 0.2 the response collapses onto its prior (divergence 3×10^{-4} per dimension) with no gain in agreement; 0.1 is chosen, with the consequence, reported in Section 3.6, that $\mathbf{w}$ is near-pinned to its prior and per-cell anomaly scores read conservatively. For ω , 0.5 loses agreement everywhere and 2 buys reconstruction at the price of more leak; $\omega = 1$. [PENDING RUNS: three seeds]

Convergence check and escalation. Refitting the closed-form operating point to convergence (best epoch 124 of a 200-epoch budget) leaves the linear probe at the floor (0.026) while the nonlinear probe reads 0.153, i.e. 46% of the converged uncontrolled baseline (0.333). The residual dependence is nonlinear, which the Gaussian penalty cannot remove by construction; the rule of Section 2.5 fires and the adversary is built. [PENDING RUNS: three seeds]

Round 3: an under-trained adversary is a placebo. With two head updates per model update at learning rate 10^{-3} and $\alpha_a \in \{0.02, 0.05, 0.1\}$, the training-time head loss looks defeated while the fresh nonlinear probe still finds 58 to 99% of the uncontrolled leak. The encoder defeats heads that are updated too slowly. Head strength must be verified by the independent probe, never by the heads' own loss. [PENDING RUNS: three seeds]

Round 4: the adversarial operating point. With six head updates per model update at learning rate 2×10^{-3} , $\alpha_a = 0.3$ brings the nonlinear probe to 14% of the uncontrolled baseline at *no* cost in type agreement (0.641 versus 0.623 uncontrolled) and about 0.02 nats per count of reconstruction; $\alpha_a = 1$ reaches the floor at agreement 0.650. The operating point is $\alpha_a = 0.3$. [PENDING RUNS: three seeds; a finer scan between 0.3 and 1]

Two facts about the weights. First, the scaling argument of Section 2.4 predicts a bound-equivalent $\alpha \approx 1/\bar{\ell}$; with a mean total of 262 counts that is 0.004, with the median 178 it is 0.006, and the $\alpha_z = 0.007$ used was set on synthetic data with about 150 counts per cell and not rescanned on the section. [PENDING RUNS: scan $\alpha_z \in \{0.004, 0.007, 0.02\}$] Second, $\alpha_w = 0.1$ is roughly 15 to 25 times the bound-equivalent value and was set by the scan above, not by the rule.

F SYNTHETIC RECOVERY

Goal. Certify, on data simulated from the model's own generative story, that the inference procedure recovers the planted intrinsic states, responses and gene programmes, that the invariance penalty keeps the planted niche out of $\mathbf{z}$, and that leakage produces the confound in the counts that the leakage sweep exists to address.

Generative process. $N = 6,000$ cells uniformly in a $1,000 \mu\text{m}$ box; $K = 8$ types assigned by a soft Voronoi partition around random anchors with Gumbel noise so that types cluster in space; the graph, kernel, composition and rings built by the same procedure as for real data from a Delaunay triangulation, with unit face lengths so that $\beta \propto \exp(-d/\tau)$ only; $\mathbf{z}_i \sim \mathcal{N}(\mathbf{m}_{t_i}, 0.5^2 \mathbf{I})$ with type means $\mathbf{m}_t \sim \mathcal{N}(\mathbf{0}, 1.5^2 \mathbf{I})$; a planted response $\mathbf{w}_i = \mathbf{M}^\top \mathbf{y}_i + 0.3 \boldsymbol{\varepsilon}_i$ linear in the true neighbour composition; $\log \boldsymbol{\rho}_i = \mathbf{A} \mathbf{z}_i + \mathbf{B} \mathbf{w}_i$ (up to normalisation) with standard-normal loadings; $\bar{\boldsymbol{\rho}}_i$ the kernel-weighted neighbour average; $\mathbf{p}_i$ the renormalised mixture at the planted κ ; totals $\ell_i \sim \max(\text{Poisson}(150), 20)$; $\mathbf{x}_i \sim \text{Multinomial}(\ell_i, \mathbf{p}_i)$; and a planted image descriptor $\Phi_i = \mathbf{P}^\top \mathbf{y}_i + 0.4 \boldsymbol{\varepsilon}'_i$ in eight dimensions. $G = 60$ genes, planted $d_z = 4$, $d_w = 2$.

Fit. A reduced model ($d_z = 8$, $d_w = 2$, hidden width 128, attention width 16) with the closed-form penalty ($\alpha_a = 0.02$), $\alpha_z = 0.007$, $\alpha_w = 0.1$, $\omega = 1$, tiles of 512 cells, 600 epochs with cosine annealing, and the planted κ given to the model. Two worlds: $\kappa = 0$, in which every pathway is identifiable, and $\kappa = 0.2$.

Readouts. Type agreement of μ_z against a reference from K -means on the leading principal components of the log-normalised counts (what the counts alone support) and against the planted $\mathbf{z}$; the first canonical correlation between μ_w and the planted response; the largest principal-angle cosine between the column spaces of the fitted and planted loadings (a random two-dimensional subspace of $\mathbb{R}^{60}$ scores about 0.3); and the within-type coefficient of determination of the centred composition on μ_z , against the uncontrolled value.

Result. **[PENDING RUNS: single seed; to be replaced]** At $\kappa = 0.2$ the type agreement is 0.62 against a count-based reference of 0.77 (the planted $\mathbf{z}$ itself gives 0.97, unreachable at 150 counts); at $\kappa = 0$ the response canonical correlation is 0.80 and the loading cosine 0.83. The loadings are seed-bistable under a planted leak: at $\kappa = 0.2$ the cosine ranges from 0.15 to 0.83 across seeds at one configuration, while at $\kappa = 0$ it is stable (0.65, 0.83). This is the confound of Section 2.2 observed in silico, and it is the reason the sweep on real data uses several seeds per grid point and the reason no gene-programme claim is made from a single run. **[PENDING RUNS: ten seeds per world; recovery as a function of the assumed κ when it is mis-specified relative to the planted one; a world with G closer to the panel size]**

Simulator checks. Leakage raises the kernel-weighted cosine between a cell’s composition and its neighbours’ (by more than 0.02 at $\kappa = 0.4$ relative to 0); the planted response is linearly predictable from composition ($R^2 > 0.7$); the mixture is exactly the renormalised convex combination and reduces to ρ for isolated cells.

Lessons that shaped the objective. Four facts were established here before any fit on real data. Under the $1/\ell$ scaling, unit weights collapse $\mathbf{z}$ (divergence to zero, agreement 0.21); the bound-equivalent weight is $\approx 1/\bar{\ell}$. The response weight is a knife-edge: near $1/\bar{\ell}$ the direct path lets $\mathbf{w}$ steal identity (agreement 0.12); at 0.1 and above the response pins to its prior. The covariance of the composition block is singular on the simplex, which is why one column is dropped and the penalty is computed on correlations. And at a moderate penalty strength the loading cosine rose from 0.56 to 0.83 while the within-type niche information in $\mathbf{z}$ fell, which is the penalty doing its job.

G VALIDATION PROTOCOLS

The analyses of Section 3.5 share one design: a target with a known allegiance, probed identically from μ_z and μ_w (posterior means of a converged model), reported per type and pooled as cell-weighted means, with reference rows and spatial-block cross-validation (Roberts et al., 2017) using the training tiles as fold groups (five folds of contiguous tiles). Random cross-validation is invalid for every spatial target here: autocorrelation leaks the answer through neighbouring cells landing on both sides of the split. Every analysis is cheap post hoc and is run at every point of the leakage sweep with seed envelopes.

Reference rows. (i) *floor*: the target permuted within type; (ii) *depth baseline*: the same probe from $\log \ell_i$ alone, because depth varies across tissue and a latent that encodes depth buys spatial skill technically (G2/M cells also carry about twice the mRNA **[NEEDS A SOURCE]**, so depth is informative even for intrinsic targets); (iii) for expression-derived targets, a *linear reference* from the leading 50 principal components of the log-normalised counts. The linear reference is not a ceiling: a nonlinear encoder can and does exceed it. **[TODO: name it consistently as “linear reference” everywhere]**

Cell-cycle phase (intrinsic). Continuous S and G2/M scores from standard gene sets (Tirosch et al., 2016, 18 S and 34 G2/M markers on the panel; split-half reliability 0.22 and 0.52); hard phase calls default to G1 at this depth and are not used. Within-type ridge regression from each latent on the four types with the highest fraction of proliferation-marker-positive cells, reported as a per-type-centred cell-weighted pooled R^2 ; the mean over types is dominated by types whose within-type score variance is near zero and is reported only in the by-type table. **[TODO: consider gating types on within-type score variance]**

Discrete niche (spatial). Niche labels from k -means on neighbour composition over connected cells (10 niches; 6 and 15 as robustness), never from the latents. Types with at least two niches of at least 500 cells are eligible; per type, a class-balanced multinomial logistic regression from each latent, reported as macro one-versus-rest area under the curve (Hand and Till, 2001) and balanced accuracy. Expectation: $\mathbf{w}$ well above the depth baseline; $\mathbf{z}$ within noise of the larger of floor and depth baseline. The observed excess of $\mathbf{z}$ over the depth baseline is reported as it is, and triaged per dimension by the Moran’s I analysis below.

Moran’s I per dimension (both latents). Moran’s I (Moran, 1950) of each dimension centred per type, so that the legitimate type mosaic is removed from both latents while the full graph is kept; the model’s own pruned graph with binary row-normalised weights; a permutation null from at least 1,000 shuffles of the centred values; dimensions whose variance is below 10^{-3} of the largest are flagged as collapsed and excluded from the mean. A high-autocorrelation $\mathbf{z}$ dimension is a flag, not a failure: intrinsically spatial biology (clone patches, proliferative niches) is allowed, since $\mathbf{z}$ claims independence of context-induced variance, not spatial uniformity. Each high-autocorrelation $\mathbf{z}$ dimension is triaged by its in-sample R^2 on composition (leakage) and its correlation with the cycle scores (benign).

Pseudotime and the tissue gradient. A principal curve (Hastie and Stuetzle, 1989) in each latent space; its agreement with the tissue gradient measured as the R^2 of the pseudotime on neighbour composition and as the graph-neighbour coherence of the pseudotime, both raw and after partialling out per-type means. The raw values are dominated by type homophily and are reported only to show it.

Distance to landmarks (spatial, geometry-derived). Landmark classes built from types and positions only: vasculature (endothelial cells, with pericytes included when they lie within $30\ \mu\text{m}$ of endothelium, clustered into instances by density-based clustering (Ester et al., 1996)), compact smooth-muscle structures (instances of 20 to 500 cells), and the tumour–stroma interface (from a k -nearest-neighbour-smoothed tumour fraction, taking the boundary). Target $\log(1 + \min(d, 500\ \mu\text{m}))$; the landmark’s own constituent type excluded from its regression; bands *near* ($< 30\ \mu\text{m}$, one hop, a composition echo and sanity check only), *mid* (50 to $300\ \mu\text{m}$, the evidence, since one-hop context cannot see the landmark), and *far* (300 to $500\ \mu\text{m}$, expected failure). Ridge regression fitted and scored within each band. Because the landmark sets are type-defined and types are expression-derived, a fourth reference row, the probe from raw neighbour composition, measures the definitional share of any signal. The probe direction per landmark is unit-normalised and compared across landmarks by cosine within one run, never across runs; gene signatures are read as $\mathbf{B}\hat{\theta}$ restricted to genes expressed in at least 1% of cells. **[TODO: the interface landmark is roughly 60% definitional and vasculature is a null on the existing run; decide whether the analysis appears beyond the appendix]**

Allegiance matrix. Rows: cell-cycle score, niche label, mid-band landmark distance, type-partialled pseudotime gradient. Columns: μ_z , μ_w , with floor, depth baseline and linear reference marked per cell. Expected pattern: block-diagonal.

H ADDITIONAL RESULTS

[TODO: Skeleton. Each subsection states what will be reported; numbers follow the reruns.]

H.1 Label-Set Control

The ovarian section refitted with an unsupervised graph-based clustering ($K = 26$) in place of the curated annotation, at the operating point: held-out reconstruction, type agreement against the new labels, the cell-cycle asymmetry, and the niche probes. **[PENDING RUNS: single fit: best held-out reconstruction -7.256 nats per count, agreement 0.70 against the 26 clusters, cell-cycle R^2 from $\mathbf{z}$ lower than under the curated labels; three seeds; interpretation of the lower cycle R^2 under finer labels, which Appendix B.4 predicts]**

H.2 Low-Degree Cells Along the Sweep

Held-out reconstruction from $\kappa = 0$ to 0.4, globally and in the strata of cells that lost at least one edge to pruning and of cells with at most two neighbours. **[PENDING RUNS: three seeds: global cost 0.030 nats per**

count, lost-edge stratum 0.07, low-degree stratum 0.10] The fixed leak fraction is a poorer model for sparsely connected cells, which is the reason they are excluded from effect readouts.

H.3 Stability of the Loadings

Mean absolute correlation between columns of $\mathbf{B}$ after optimal matching (Kuhn, 1955), across seeds within each κ and along κ within a seed lineage. **[PENDING RUNS: three seeds: across-seed value about 0.62 to 0.64 for $\kappa \leq 0.1$ and 0.48 to 0.55 for $\kappa \geq 0.2$; along- κ within one seed 0.83 to 0.93]** The loadings are therefore identified well enough to track a programme along the sweep within a run and not well enough to name a programme from one run; all gene-level statements are run-relative (Section 2.8). **[PENDING RUNS: five seeds]**

H.4 Per-Type Response Magnitude

Mean $\|\mu_w\|$ per type on held-out cells, at every κ and seed. **[PENDING RUNS: three seeds: the ordering of the top three types (VEGFA-positive tumour, proliferative tumour, inflammatory tumour) holds in 16 of 18 runs and the top-three set in 17 of 18]** The exceptions are reported, not dropped.

H.5 The Response at the Operating Point

[PENDING RUNS: one converged reference fit] μ_w is explained by the prior mean $m_\psi(\mathbf{c}, t)$ with $R^2 = 0.9998$, the per-dimension divergence is 2×10^{-4} to 1.5×10^{-3} , and the first two principal components of μ_w carry 85% and 15% of its variance (participation ratio 1.3). At this operating point $\mathbf{w}$ is, to a very good approximation, a low-dimensional deterministic function of type and context; the allegiance analyses therefore test the context channel as a whole, which is still the allegiance claim, and the per-cell anomaly score is deactivated. **[PENDING RUNS: three seeds; a lower- α_w arm for comparison]**

H.6 Run Inventory

[TODO: table of every fit that contributes a number to the paper: section, labels, κ , seed, epochs to best, wall time.]

Table 2: Networks and fixed quantities. G genes, K types, d_Φ image-embedding dimension.

Component	Input	Layers / value
Count encoding $q(\mathbf{z}_i \mathbf{x}_i, t_i)$	$\mathbf{x}_i$ count encoding $\oplus$ onehot(t_i)	$[\log(1 + \mathbf{x}_i \tilde{\ell} / \ell_i), \log \ell_i]$, $\tilde{\ell}$ the median total $256 \rightarrow 256 \rightarrow 2d_z$, $d_z = 20$
Type query embed(t_i)	t_i	learned table, width $K + d_z$
Attention sources	neighbour j	$[\text{onehot}(t_j), \text{sg}[\mu_z(\mathbf{x}_j)]]$, width $K + d_z$
Attention layer	sources, query	one bipartite GATv2 layer, 4 heads $\times$ 32, averaged, LeakyReLU slope 0.2, no self-loops, no edge features
Context $\mathbf{c}_i$		attention output (32) $\oplus \Phi_i$ ($d_\Phi = 384$) $\oplus$ isolation flag (1)
Prior mean $m_\psi(\mathbf{c}_i, t_i)$	$\mathbf{c}_i \oplus \text{onehot}(t_i)$	$64 \rightarrow d_w$, $d_w = 6$
$q(\mathbf{w}_i \cdot)$	$\mathbf{c}_i \oplus \text{onehot}(t_i) \oplus \mathbf{z}_i \oplus$ count encoding	$256 \rightarrow 2d_w$
Decoder $a(\mathbf{z}_i)$	$\mathbf{z}_i$	$256 \rightarrow G$
Loadings $\mathbf{B}$	$\mathbf{w}_i$	linear $d_w \rightarrow G$, no bias
Adversary heads	$\mu_{z,i} \oplus \text{onehot}(t_i)$	two heads $64 \rightarrow 64 \rightarrow K$ (composition) and $\rightarrow E_\Phi = K$ (image niche)
Image niche e_i^Φ	12 principal components of Φ	soft k -means memberships $\text{softmax}(-d^2/T)$, T the median nearest squared distance; the projection is fitted on a 50,000-cell random subsample and applied to all cells, the k -means (four restarts) on all cells; fitted once and frozen
Invariance block $\mathbf{v}_i$		$[\mathbf{y}_i^{(-K)}, 12 \text{ principal components of } \Phi_i]$
Held-out probes	$\mu_{z,i} \oplus \text{onehot}(t_i)$	ridge regression ($\lambda = 10^{-3}$) with a within-type permutation floor; perceptron $64 \rightarrow 64$, 300 Adam steps at 10^{-3} with batches of 2,048; both fitted on at most 30,000 training-tile cells
Graph		Delaunay adjacency; Voronoi faces measured within $30 \mu\text{m}$ discs; prune at $40 \mu\text{m}$
Kernel β		$f_{ij} \exp(-d_{ij}/\tau)$, $\tau = 20 \mu\text{m}$, row-normalised over in-edges
Prior scale σ_w		1 (fixed)
Leak fraction κ		swept over $\{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$; operating point 0.1
ω		1
α_z		0.007
α_w		0.1
α_a (adversary)		0.3; heads updated 6 times per model update, learning rate 2×10^{-3}
α_a (closed form, not used for the sweep)		0.03, calibrated (Appendix E)
Moving-average rate, shrinkage		0.05, 0.05; support floor $10 \times$ joint dimension
Tiles		recursive median split into rectangles of $\leq 4,096$ cells; 15% of tiles held out
Optimiser		Adam (Kingma and Ba, 2015), learning rate 10^{-3} , cosine annealing (Loshchilov and Hutter, 2017) over 200 epochs, gradient-norm clip 10
Evaluation		every 5 epochs on posterior means; checkpoint accepted if held-out reconstruction improves and $\mathbf{z}$ -type agreement $\geq 0.9 \times$ its running maximum; patience 20 epochs
Halo overhead		extra nodes per step 21% of the seed count at tiles of about 1,600 cells and 9.7% at about 6,400 (not measured at 4,096)
Hardware		one consumer GPU; a 407,120-cell fit takes roughly 7 s per epoch, 7 to 23 minutes to early stopping

Table 3: Implementation choices and their reasons.

Where	Decision	Reason
Type input to $q(\mathbf{w})$ and m_ψ	one-hot everywhere the type is an input; the learned type embedding exists only as the attention query, sized to match the source features	the query stands in for the $\mathbf{z}_i$ that the cell deliberately does not contribute to its own context
Counts input to $q(\mathbf{w})$	the same encoding as $q(\mathbf{z})$	one encoding of the counts for both posteriors
Neighbour responses $\mathbf{w}_j$	a posterior draw from the same batched pass, under a stop-gradient	the neighbours' clean compositions must include their local programme (Section 2.6)
Invariance block	one composition column dropped	the simplex makes the full block singular; the information is unchanged (Appendix A.4)
Penalty matrices	correlation rather than covariance form, plus $10^{-5}\mathbf{I}$ on the correlation matrix before the log-determinant	identical value by scale invariance; bounded gradient; on a unit-diagonal matrix the additive term is a uniform relative jitter, not a variance-inflating ridge
Penalty gradient	straight-through: value from the moving averages, gradient from the batch	otherwise the effective strength of α_a scales with the averaging rate
Penalty, adversary and probe input	posterior means $\boldsymbol{\mu}_z$, not samples	encoder noise would dilute the measured dependence
Objective granularity	per-cell mean over the batch rather than a sum	invariant to tile size
Evaluation draws	posterior means	the stopping signal should not ride reparameterisation noise
Image embedding dimension	full 384 dimensions into $\mathbf{c}_i$ rather than a 32-64-dimensional projection	an ablation found a small gain in held-out reconstruction from the full embedding (Appendix E); the invariance block still uses 12 principal components
Probe target for the image	Gaussian cross-entropy on the continuous block rather than categorical cross-entropy against image-niche clusters	the closed-form penalty takes continuous components directly; a small nonlinear probe is fitted alongside as a cross-check
Graph construction	the Voronoi tessellation is clipped to $30\mu\text{m}$ discs before faces are measured, then edges are pruned at $40\mu\text{m}$	the clip removes faces only for pairs more than roughly $60\mu\text{m}$ apart, which the prune removes anyway, and bounds the length of faces at tissue borders and across gaps; the effect on β for kept edges has not been quantified [TODO: quantify]
Invariance mechanism	the adversary is the operating configuration; the closed-form penalty is the cheaper first attempt	a nonlinear probe found residual dependence after the closed-form penalty converged (Appendix E)
Not built	Dirichlet-multinomial likelihood; counterfactual machinery with abduction; bounded conditional $\sigma_w(\mathbf{c}, t)$; estimation of κ from a nuclear/extranuclear split; edge features in the attention layer	each has a stated trigger that has not fired, or is future work